Diagnostic validity and clinical utility of genetic testing for hypertrophic cardiomyopathy – A systematic review and meta-analysis

Christian et al.

Supplementary Materials

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Supplementary Methods Table 1: PICOTS for individuals with known or suspected diagnosis of HCM or for at-risk relatives of those patients.

	Genetic Testing		Genetic Counseling
	Diagnostic Validity	**Clinical Utility**	**Clinical Utility**
Population	– Patients with a known or suspected diagnosis of HCM (proband) – At-risk relatives of probands	– Patients with a known or suspected diagnosis of HCM (proband) – At-risk relatives of probands	– Patients with a known or suspected diagnosis of HCM (proband) – At-risk relatives of probands
Intervention	– Genetic testing by any method	– Genetic testing by any method	– Genetic counseling
Comparison	– n/a	**Probands** – No genetic testing – Positive results vs VUS vs non-informative result – Assessment before/after genetic testing **At-risk relatives** – No genetic testing - no cardiac evaluation – No genetic testing - periodic evaluation by cardiology	**Probands** – No genetic counseling – Education or information provided by a non-genetics specialist – Assessment before/after genetic counseling **At-risk relatives** – No genetic counseling – Education or information provided by a non-genetics specialist – Assessment before/after genetic counseling
Outcomes	**Probands** – Diagnostic yield – Inconclusive/VUS rate – Frequency of 2 pathogenic variants – Genotype-phenotype correlation **At-risk relatives** – Penetrance – Genotype-phenotype correlation	**Probands** – Change in management, behavior – Patient-reported outcomes **At-risk relatives** – Uptake of cardiac screening – Change in management, behavior – Prognosis – Patient-reported outcomes	**Probands** – Change in management, behavior – Patient-reported outcomes **At-risk relatives** – Uptake of cardiac screening – Change in management, behavior – Prognosis – Patient-reported outcomes
Timing	– n/a	– unspecified	– unspecified
Setting	– n/a	– any clinical setting	– any clinical setting

Supplementary Methods Table 2: MEDLINE search strategy for PubMed.

Query	All References	English References
(hypertroph* [tiab] AND cardiomyopath* [tiab]) OR hypertrophic cardiomyopathy[tiab] OR HCM[tiab] OR HOCM[tiab] OR obstructive cardiomyopathy[tiab] OR non-obstructive cardiomyopathy[tiab] OR nonobstructive cardiomyopathy[tiab] OR IHSS[tiab] OR idiopathic hypertrophic subaortic stenosis[tiab] OR "cardiomyopathy, hypertrophic"[mh]	20,722	16,823
mybpc3[tiab] OR myh7[tiab] OR TNNT2[tiab] OR TNNI3[tiab] OR TPM1[tiab]	1,043	991
genetic test[tiab] OR genetic tested[tiab] OR genetic testing[tiab] OR genetic testings[tiab] OR genetic tests[tiab] OR genetic counsel[tiab] OR genetic counselees[tiab] OR genetic counseling[tiab] OR genetic counselling[tiab] OR genetic counsellor[tiab] OR genetic counsellors[tiab] OR genetic counselor[tiab] OR genetic counselors[tiab] OR molecular test[tiab] OR molecular testing[tiab] OR molecular tests[tiab] OR Molecular analysis[tiab] OR Genetic analysis[tiab] OR genetic screen[tiab] OR genetic screening[tiab] OR genetic screenings[tiab] OR genetic screens[tiab] OR "genetic testing"[mh] OR "genetic counseling"[mh] OR "genetic services"[mh]	112,534	103,805
#2 OR #3	113,426	104,651
#1 AND #4	1,120	1,015

Supplementary Methods Table 3. Inclusion and exclusion criteria for literature review.

Inclusion	Exclusion
Original research of any study design (e.g., randomized controlled trial, observational study, case series)	Review, commentary, editorial, conference abstract
Adult or pediatric population	Animal or in vitro study
Patients with HCM or suspected HCM	Case reports/series N<10
Family members of patients with HCM or suspected HCM	Population not of interest (not HCM or syndromic HCM)
Reported change in patient (or at-risk family member) management, behavior, patient-reported outcomes, diagnostic yield, variant of uncertain significance rate, frequency of multiple pathogenic variants, genotype-phenotype correlation, risk assessment	No outcome of interest (e.g., only discusses treatment, management, or clinical trial)
Genetic counseling or genetic testing was performed	No intervention of interest (i.e., no genetic counseling or genetic testing)
≥10 participants	Non-English language
	No full text article available
	Duplicate article

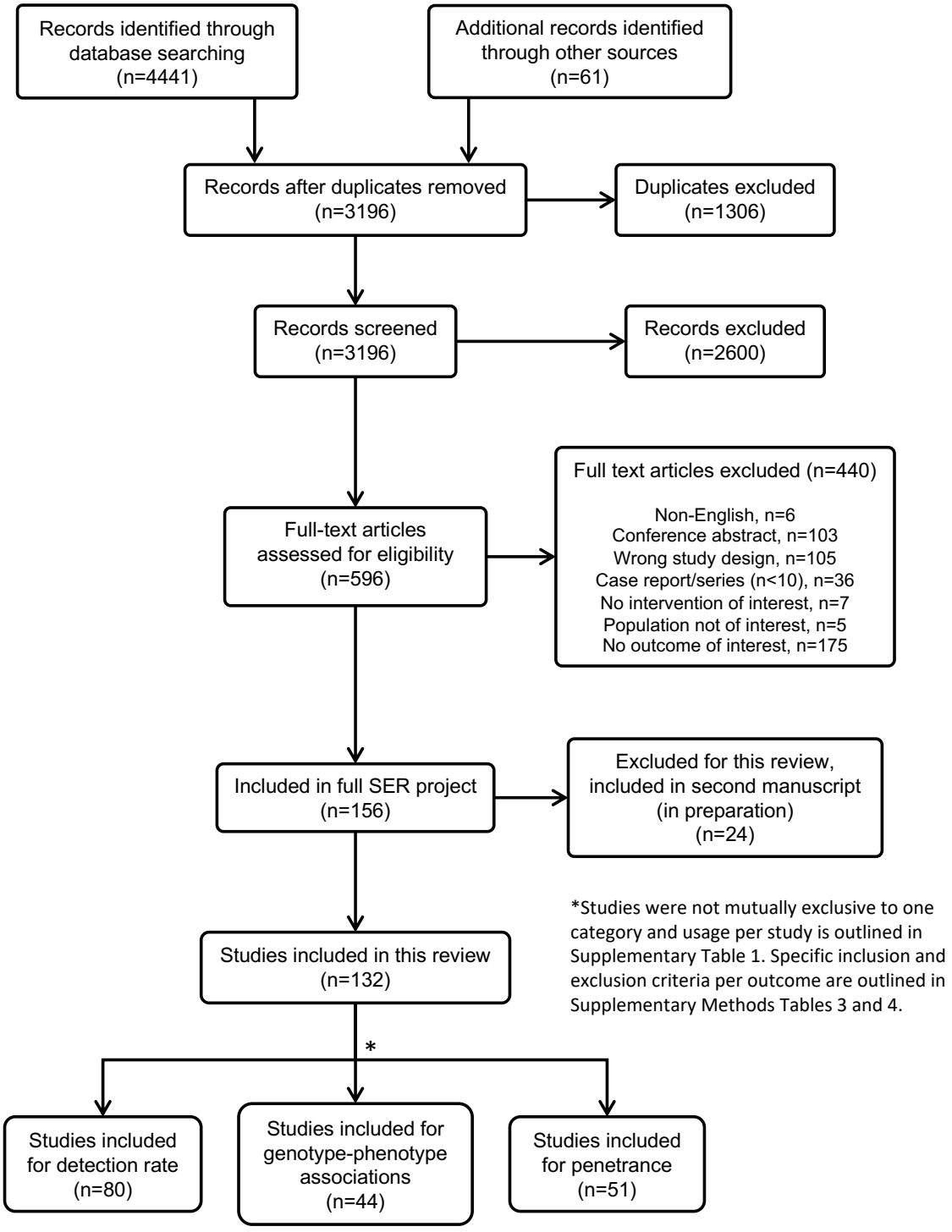
Supplementary Methods Table 4: Outcome specific exclusion criteria.

Outcome	Exclusion criteria
Detection rate	– Sequencing of <8 definitively associated HCM genes[1]
Genotype-phenotype association	– No comparison group of interest (G+/G-; MYBPC3/MYH7; multiple/single variant(s) – No genotype-phenotype data – <10 unrelated patients in each comparison group – Sequencing of <8 definitively associated HCM genes[1] (G+/G- comparisons only)
Penetrance	– <2 families evaluated

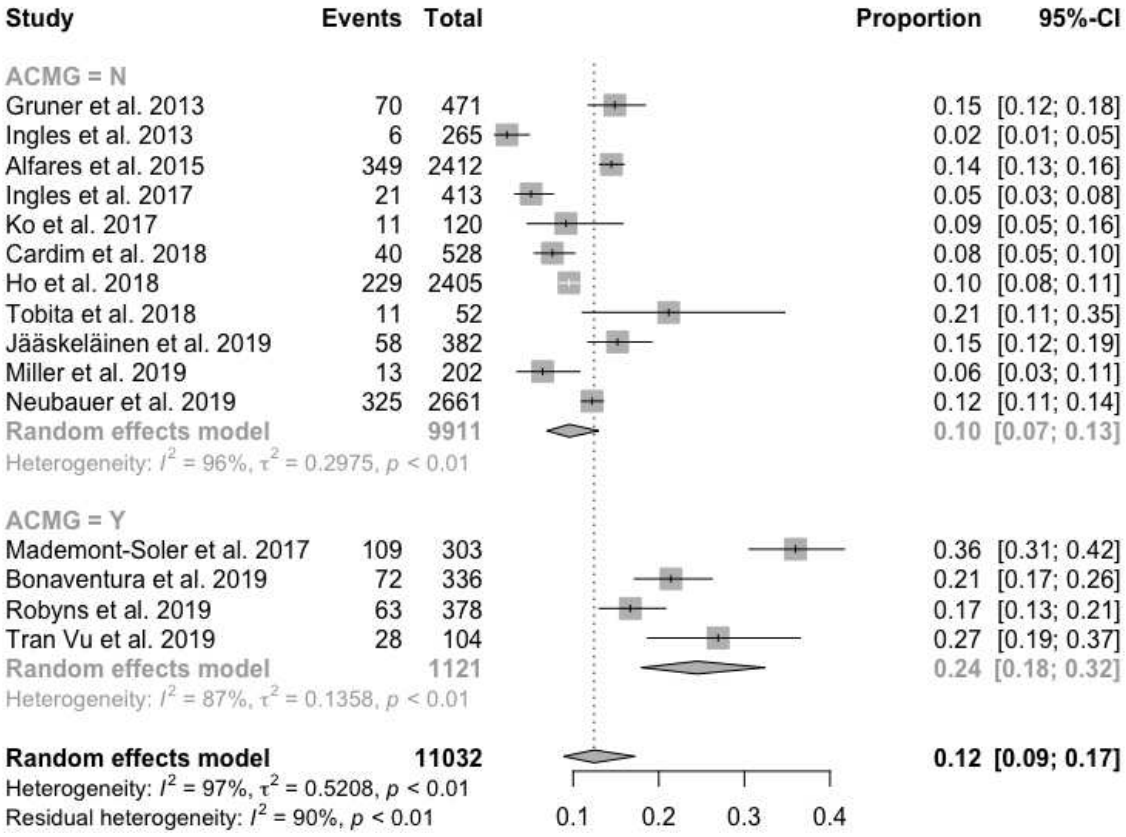
Supplemental Methods References

1. Ingles J, Goldstein J, Thaxton C, et al. Evaluating the Clinical Validity of Hypertrophic Cardiomyopathy Genes. *Circ Genom Precis Med.* 2019;12(2):e002460.

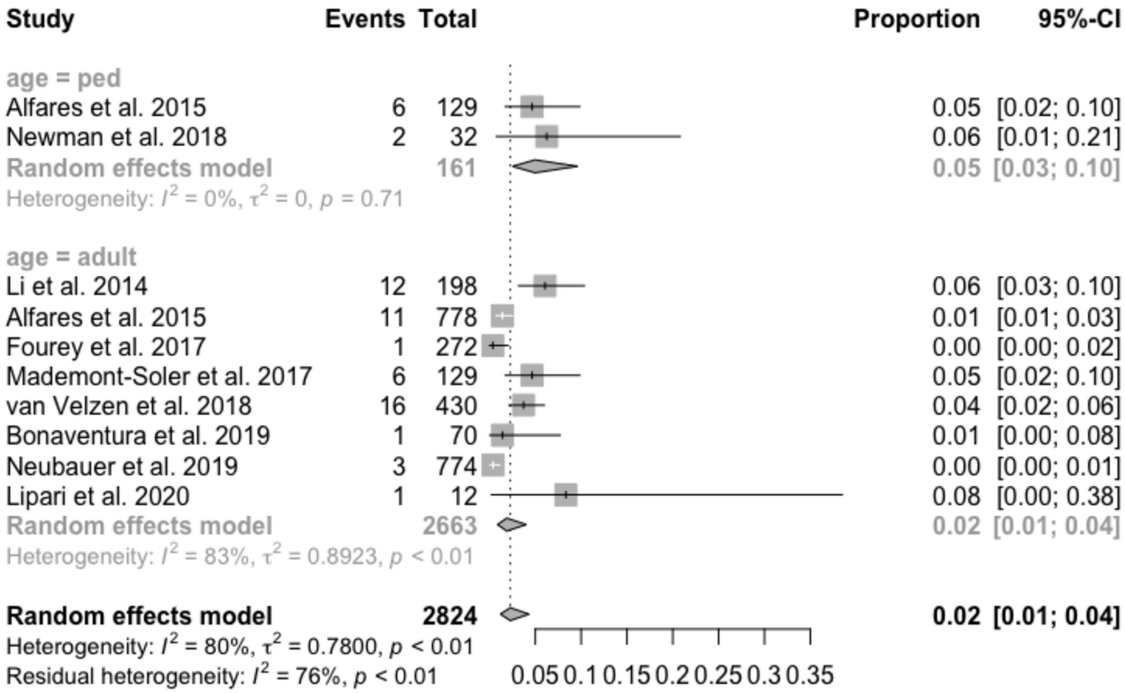
Supplementary Figure 1: PRISMA flow chart of records exclusion.



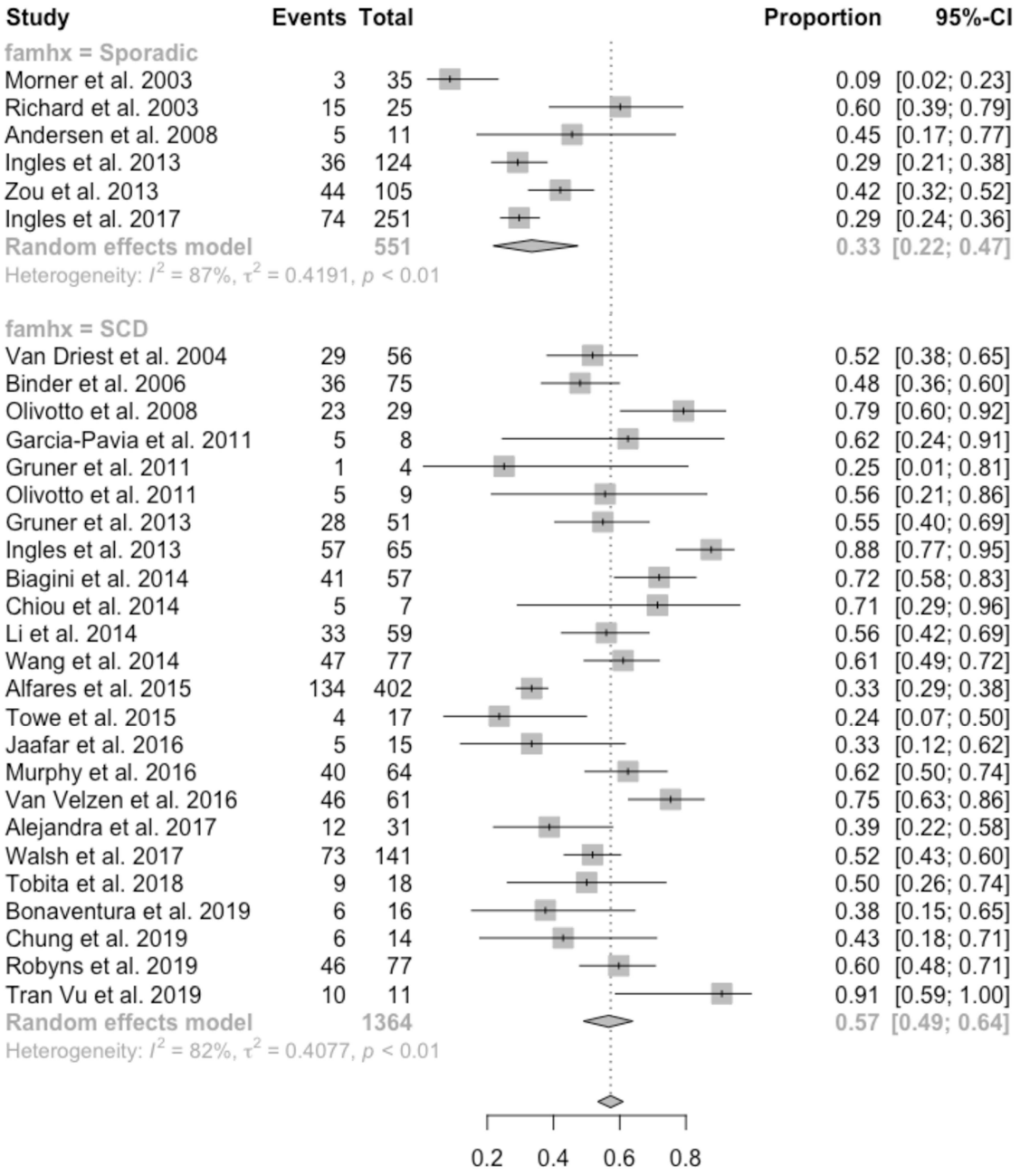
Supplementary Figure 2: Forest plot of inconclusive rate in predominantly adult cohorts with HCM by usage of the ACMG/AMP standards. Between-group comparison: p<0.0001.



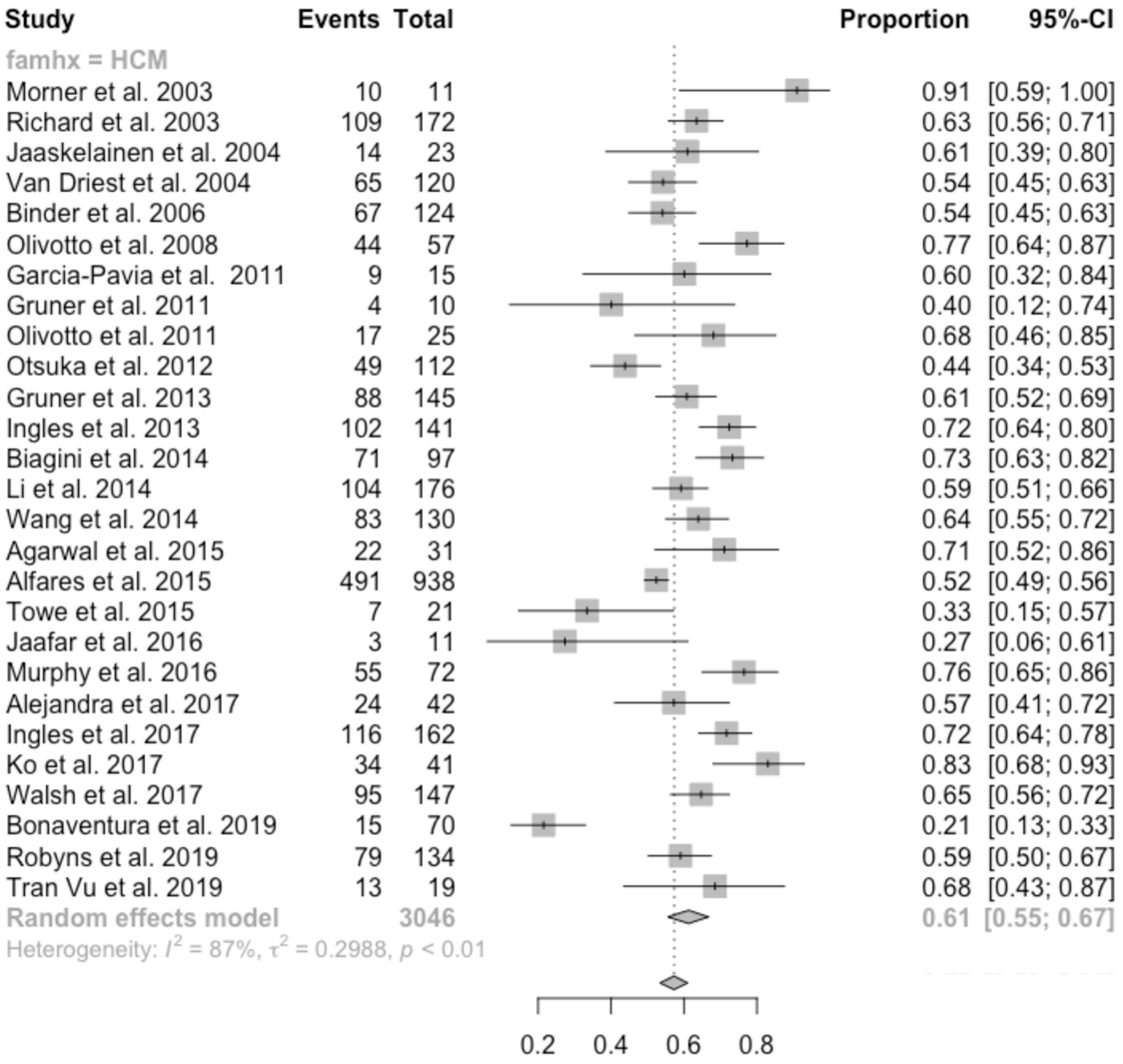
Supplementary Figure 3: Forest plot of pediatric versus adult detection rate of multiple disease-causing variants. Between-group comparison: p = 0.06.



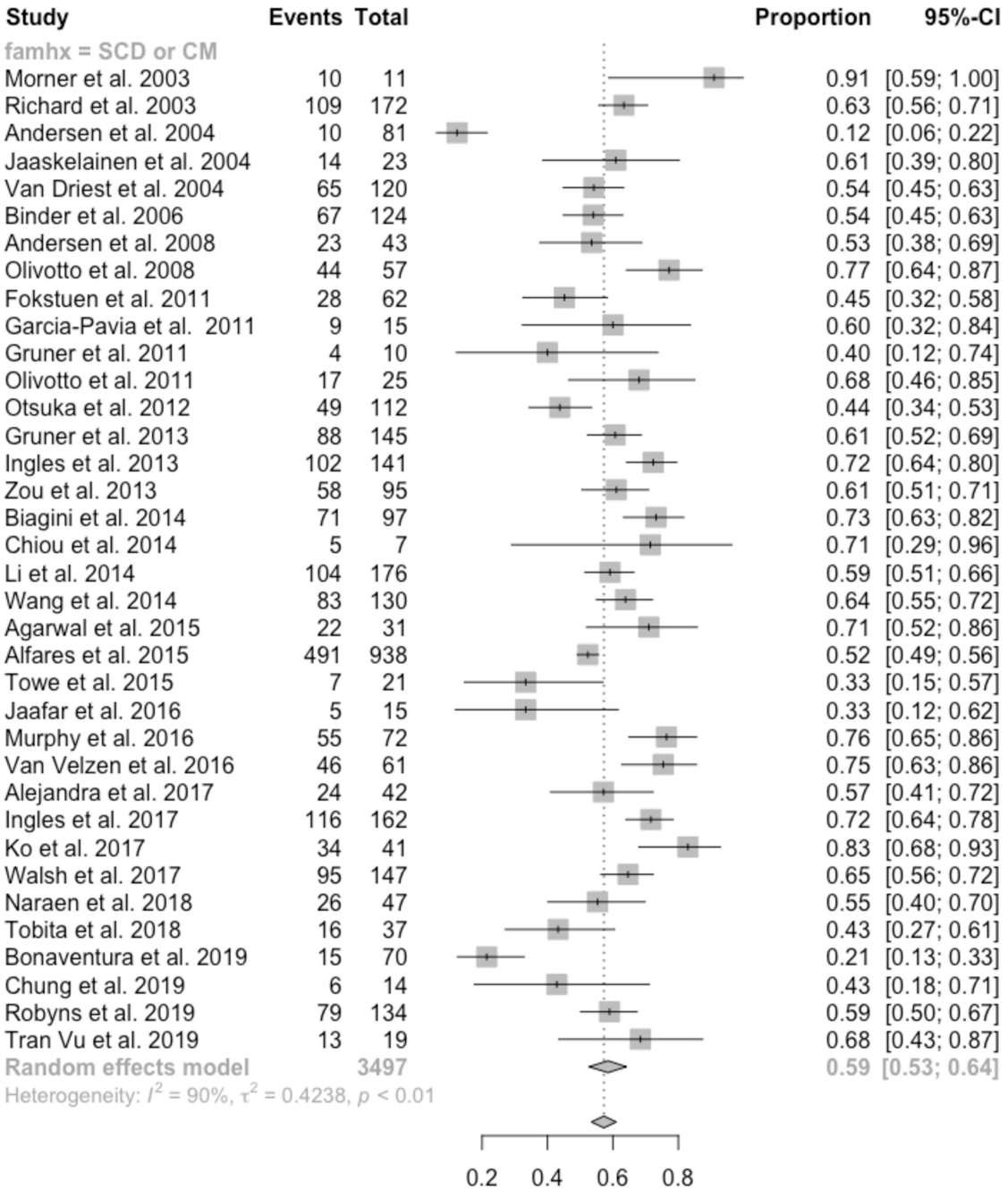
Supplementary Figure 4: Forest plot of detection rate in predominantly adult HCM cohorts by family history. Between-group comparison: p = 0.005



Supplementary Figure 4 (cont.): Forest plot of detection rate in predominantly adult HCM cohorts by family history. Between-group comparison: p = 0.005

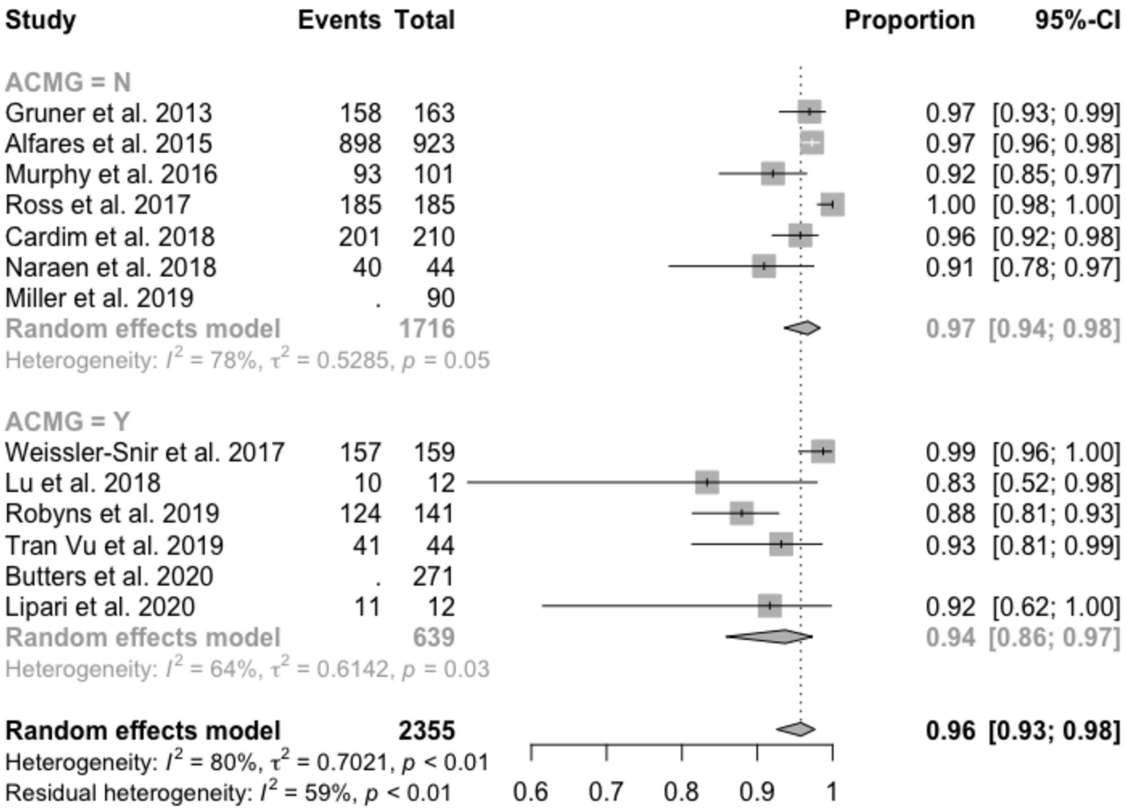


Supplementary Figure 4 (cont.): Forest plot of detection rate in predominantly adult HCM cohorts by family history. Between-group comparison: p = 0.005



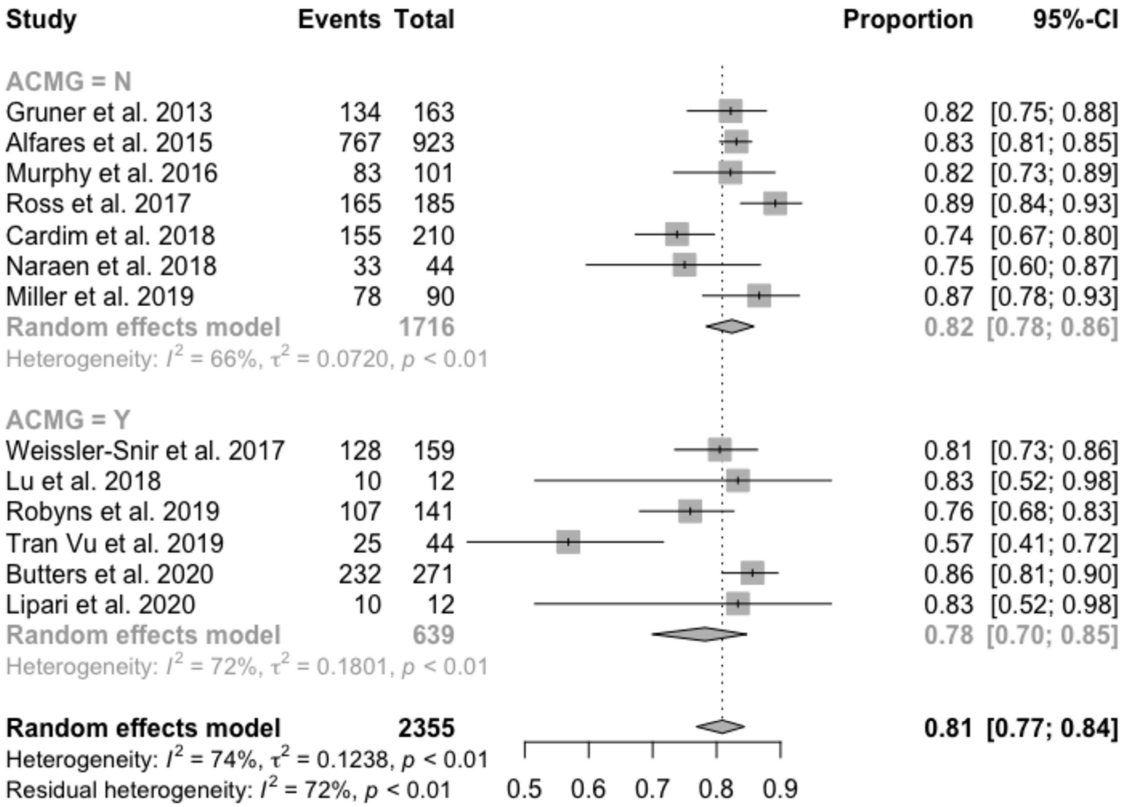
Supplementary Figure 5: Forest plot of disease-causing variants identified in the 8 sarcomere genes (A) and *MYBPC3* and *MYH7* (B).

A. 8 sarcomere genes

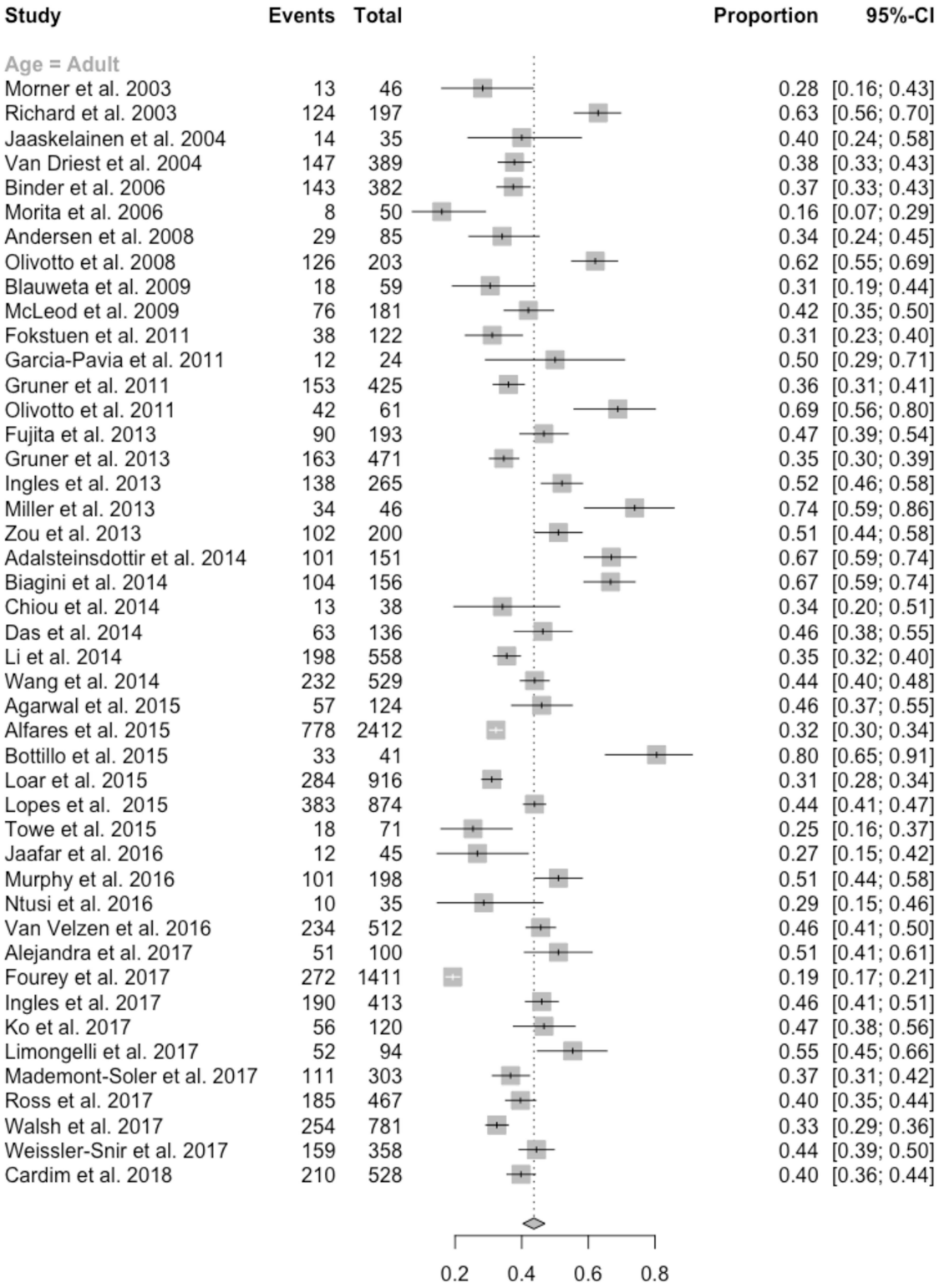


Supplementary Figure 5 (cont.): Forest plot of disease-causing variants identified in the 8 sarcomere genes (A) and *MYBPC3* and *MYH7* (B).

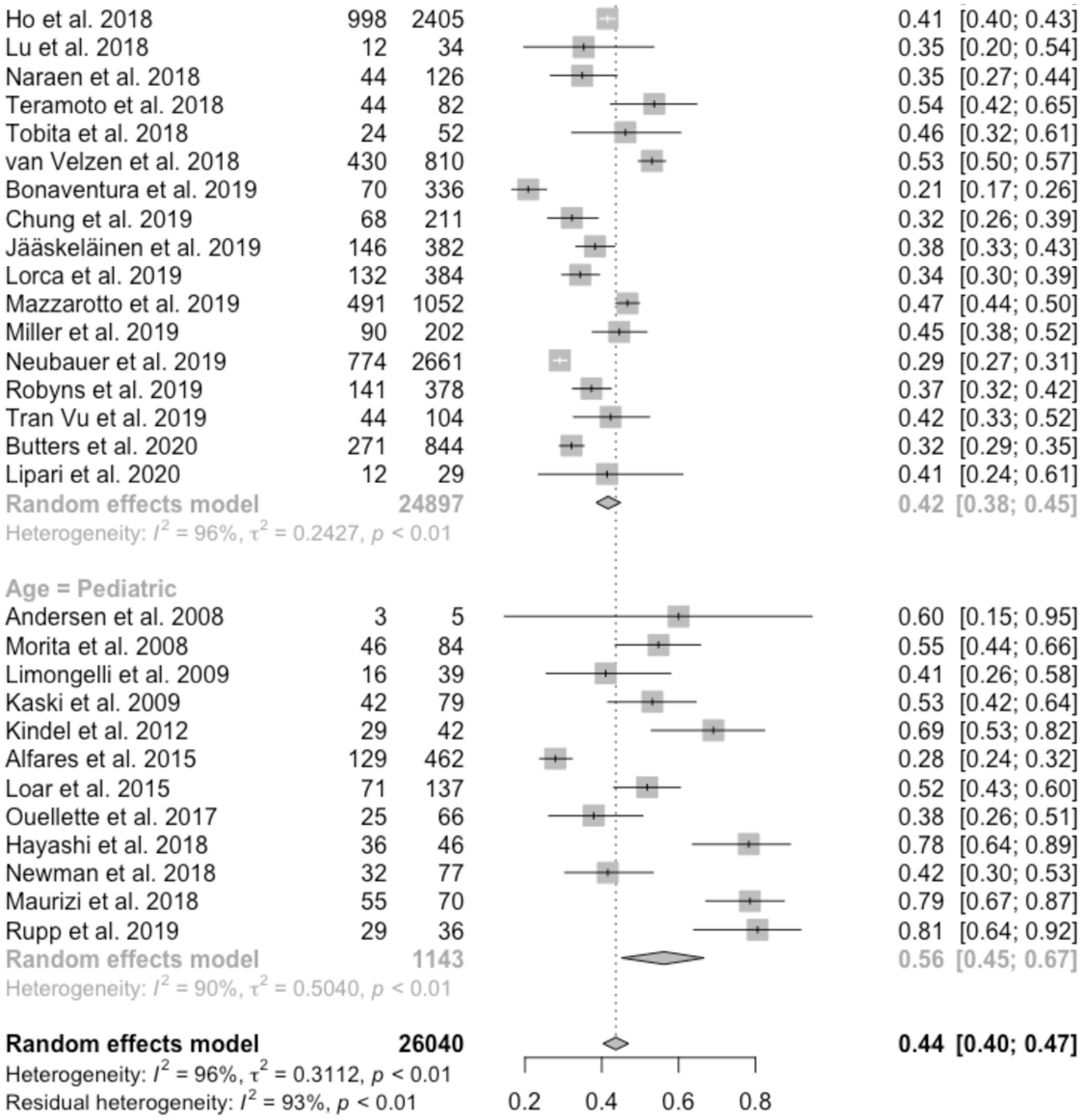
B. *MYBP3* and *MYH7* genes



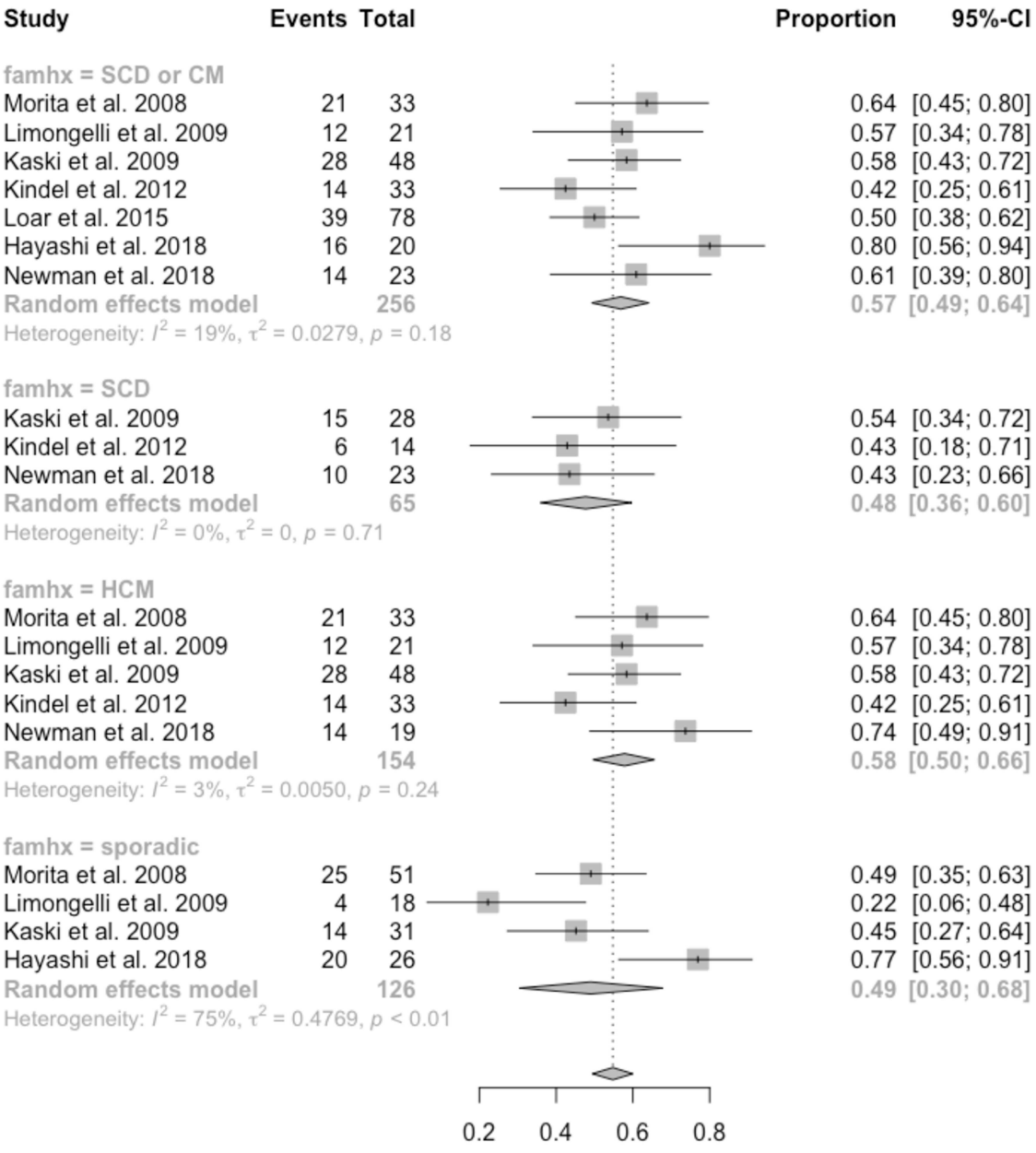
Supplementary Figure 6: Forest plot of adult versus pediatric detection rates. Between-group comparison: p=0.01.



Supplementary Figure 6 (cont.): Forest plot of adult versus pediatric detection rates. Between-group comparison: p=0.01.

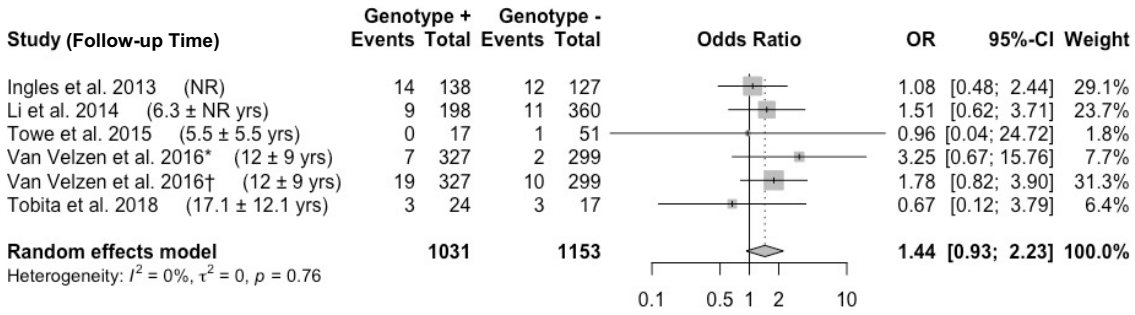


Supplementary Figure 7: Forest plot of pediatric detection rate by family history. Between-group comparison: p=0.49.

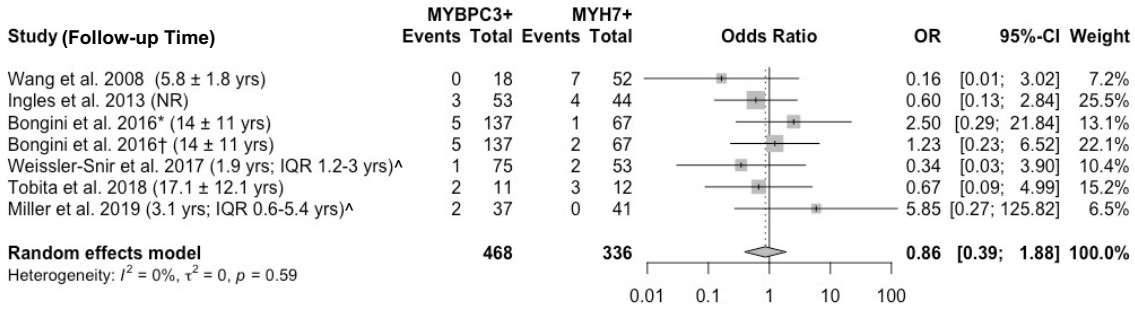


Supplementary Figure 8: Forest plots of sudden cardiac arrest (SCA), as defined in the manuscript as resuscitated cardiac arrest, SCD, appropriate ICD therapy, or a combination of these events, in predominantly adult cohorts across (A) genotype-positive (G+) versus genotype-negative (G-) cohorts and (B) *MYBPC3* versus *MYH7* cohorts. The mean follow-up time is provided in parentheses for each publication. All values are years (yrs) and NR is not reported. *Sudden cardiac death/arrest only (SCD/A). †Combined SCA/SCD/ICD shock. IQR: interquartile range. ^Reported as median rather than mean.

A. G+ vs G-

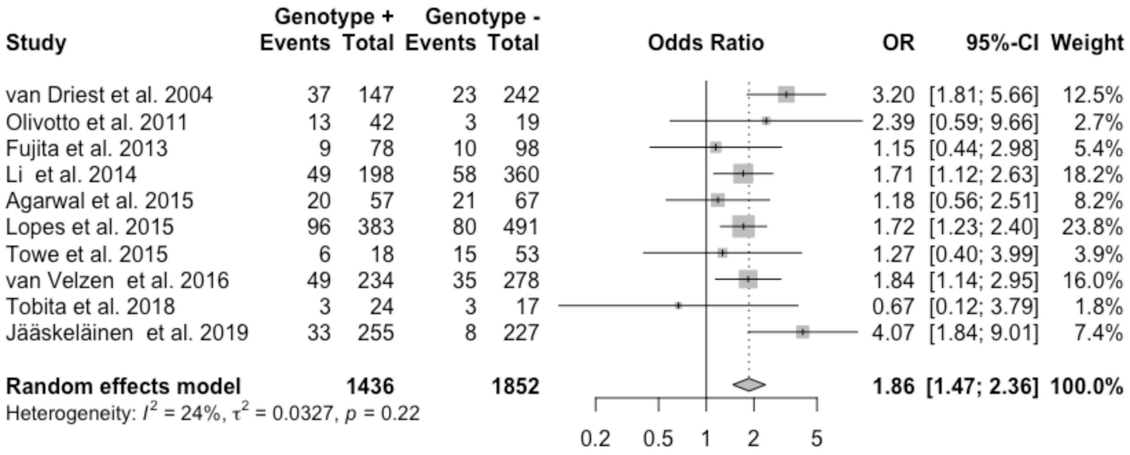


B. *MYBPC3* vs *MYH7*

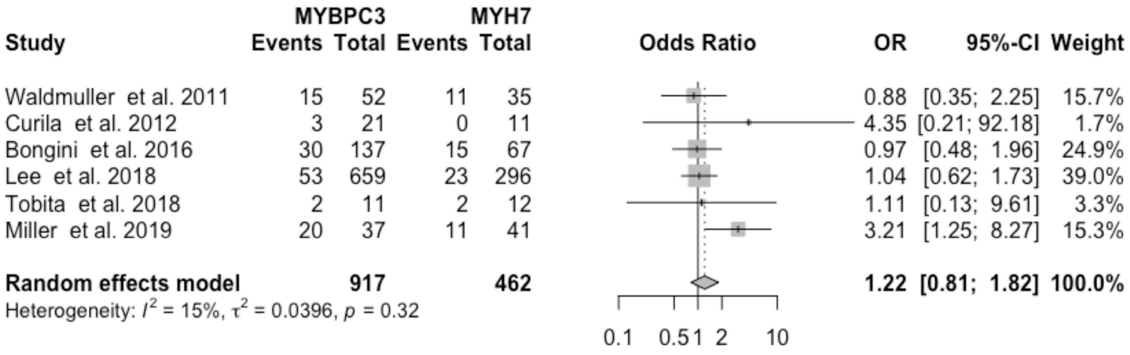


Supplementary Figure 9: Forest plots of implantable cardioverter defibrillator (ICD) placement in predominantly adult cohorts across (A) genotype-positive (G+) versus genotype-negative (G-) cohorts and (B) MYBPC3 versus MYH7 cohorts.

A. G+ vs G-

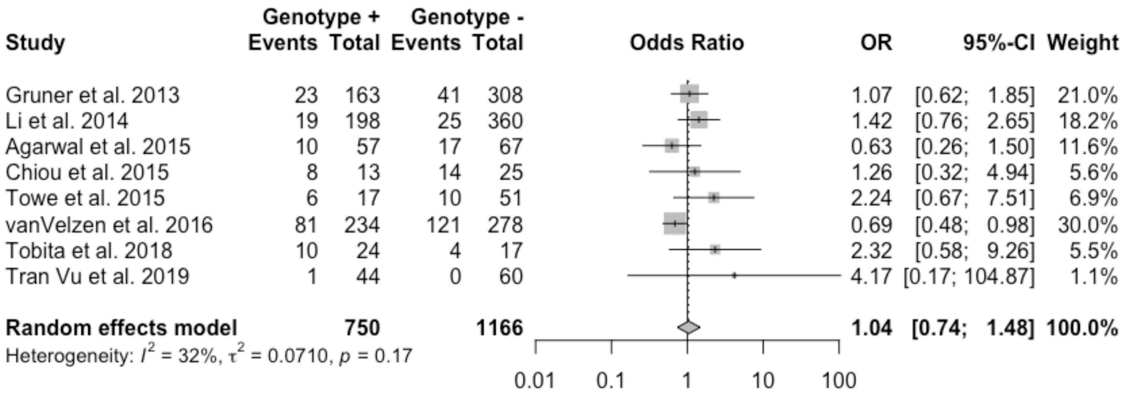


B. *MYBPC3* vs *MYH7*

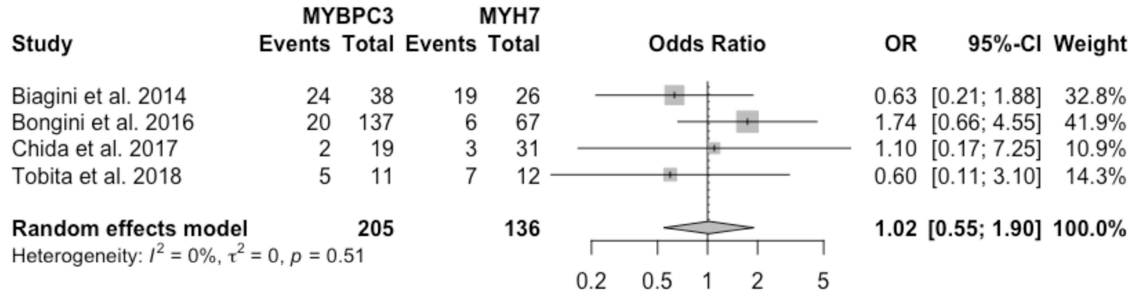


Supplementary Figure 10: Forest plot of prevalence of heart failure across (A) genotype positive (G+) and genotype negative (G-) cohorts and (B) *MYBPC3* and *MYH7* cohorts.

A. G+ vs G-

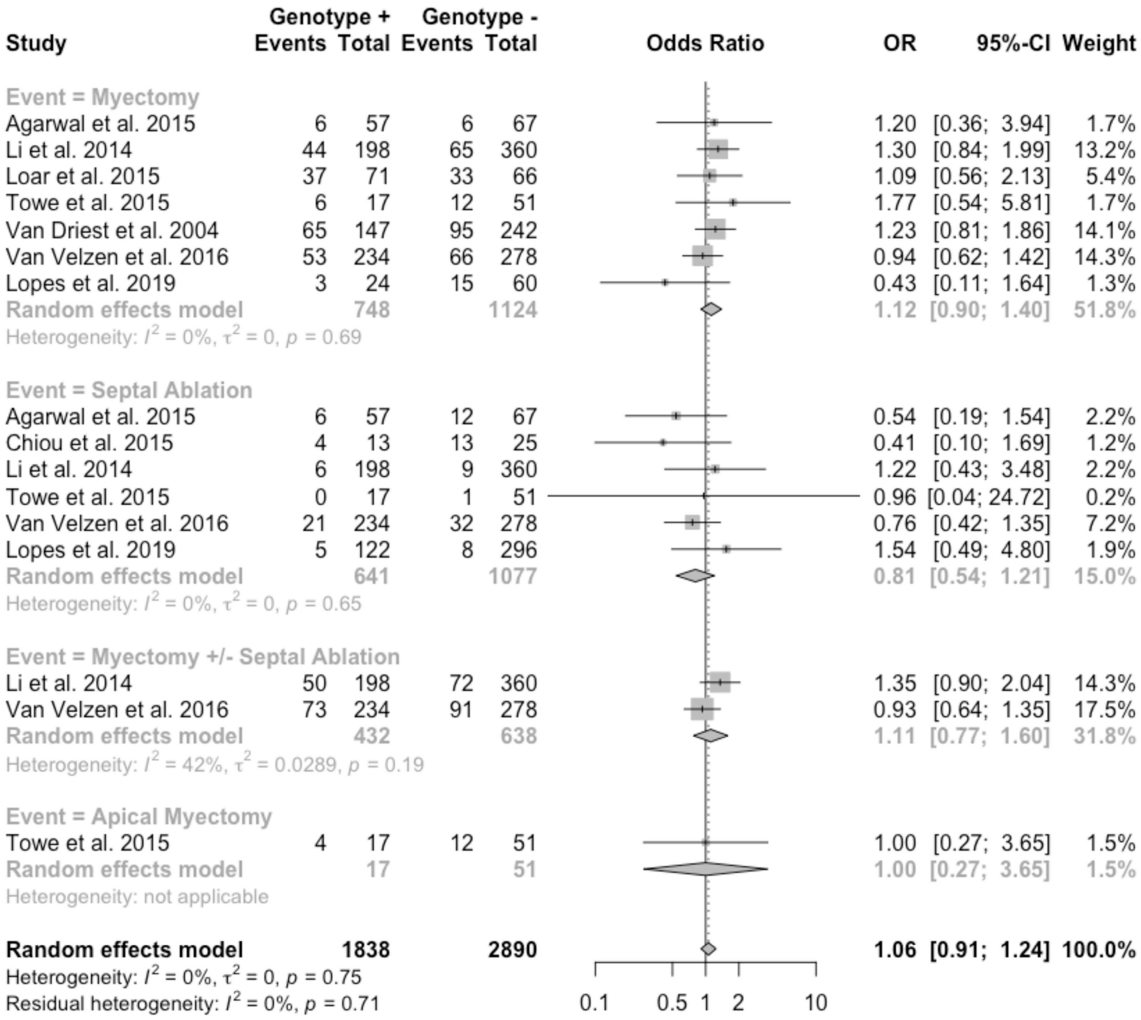


B. *MYBPC3* vs *MYH7*



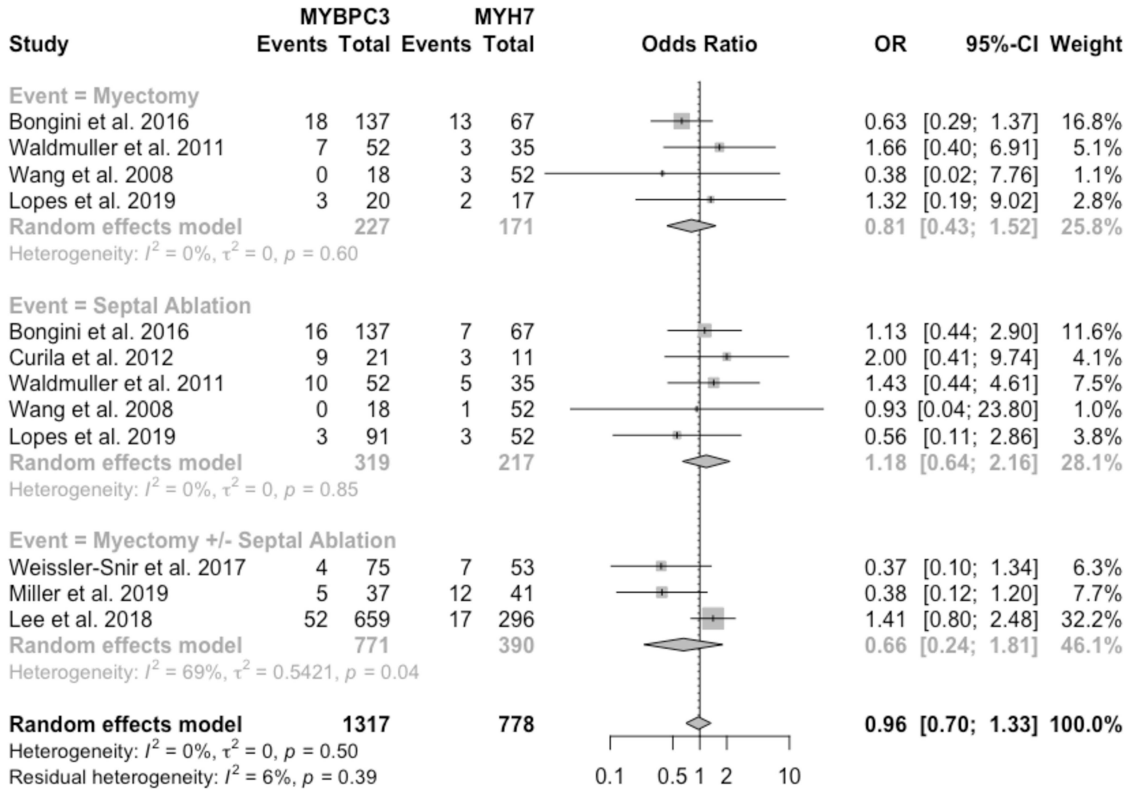
Supplementary Figure 11: Forest plot of septal reduction across (A) genotype positive (G+) and genotype negative (G-) cohorts and (B) *MYBPC3* and *MYH7* cohorts

A. G+ vs G-

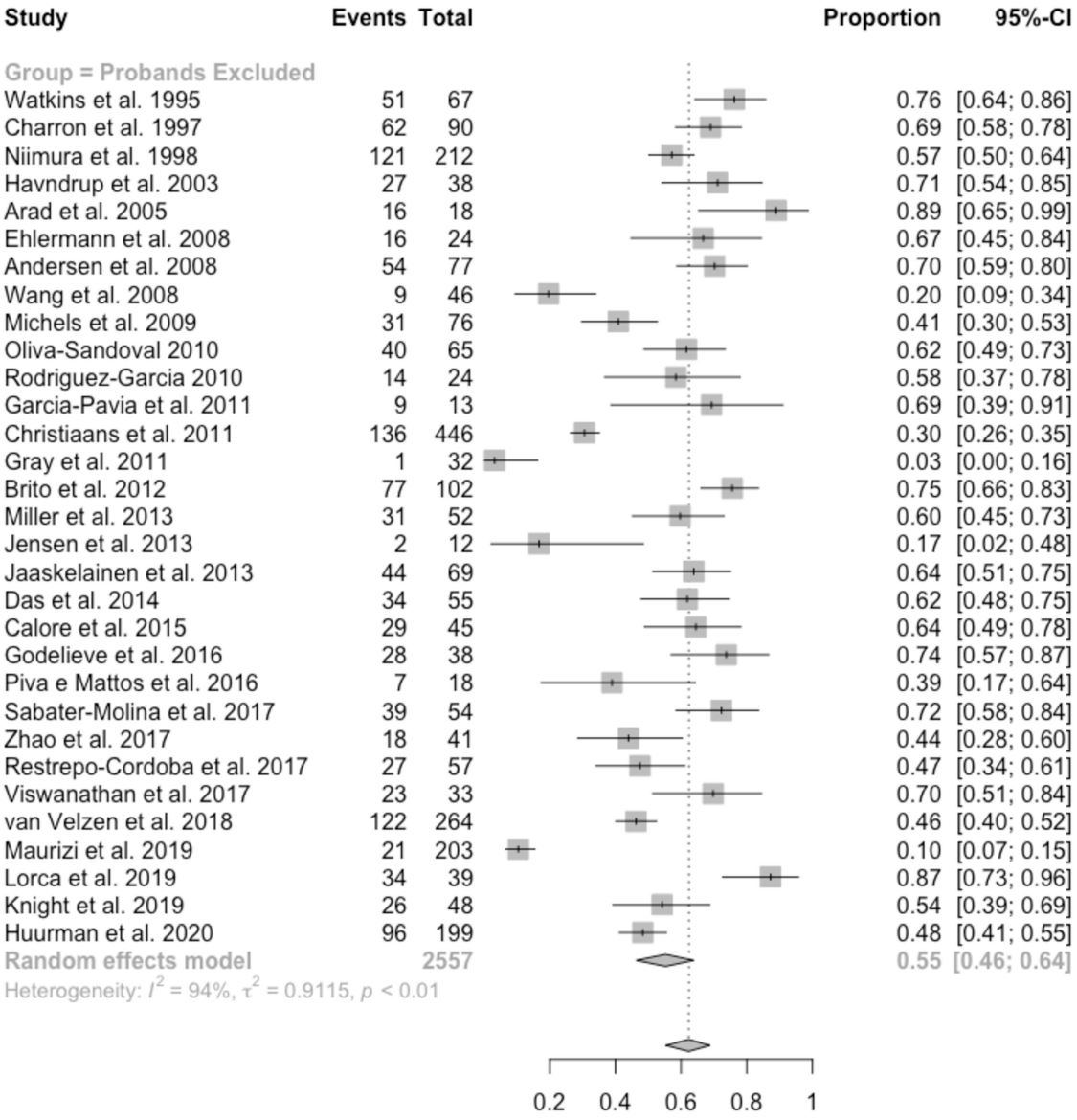


Supplementary Figure 11 (cont.): Forest plot of septal reduction across (A) genotype positive (G+) and genotype negative (G-) cohorts and (B) *MYBPC3* and *MYH7* cohorts

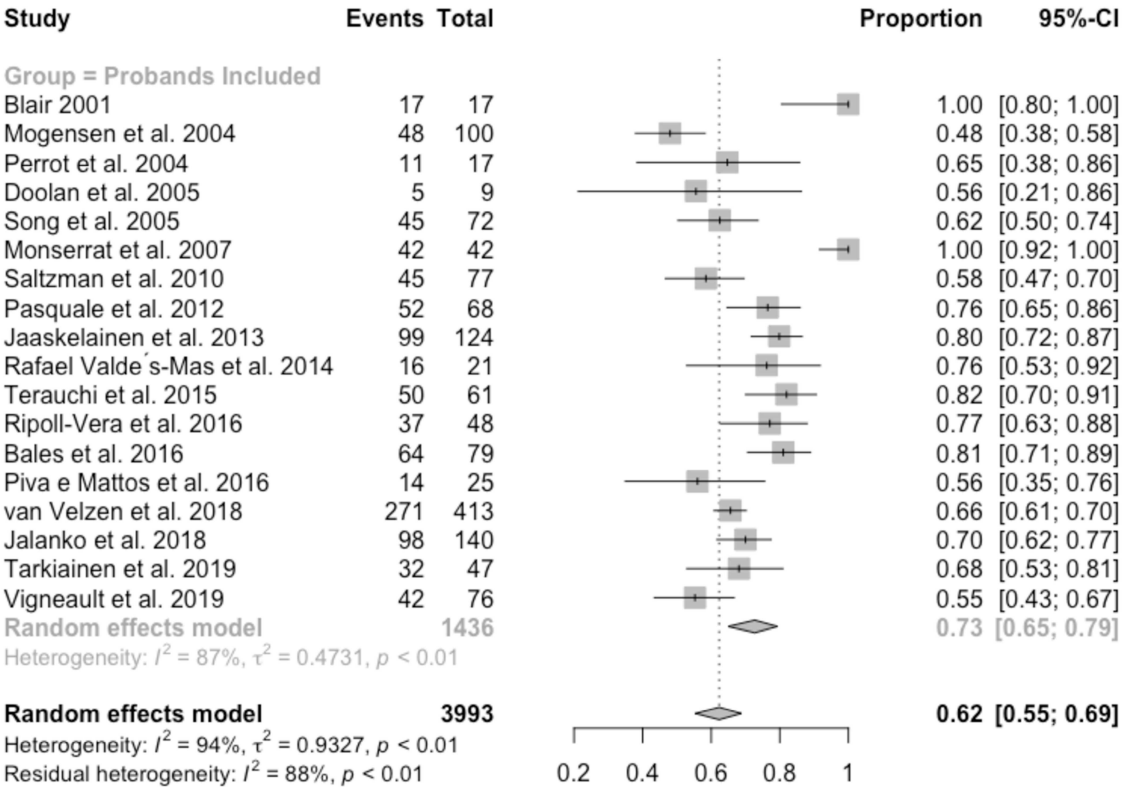
B. *MYBPC3* vs *MYH7*



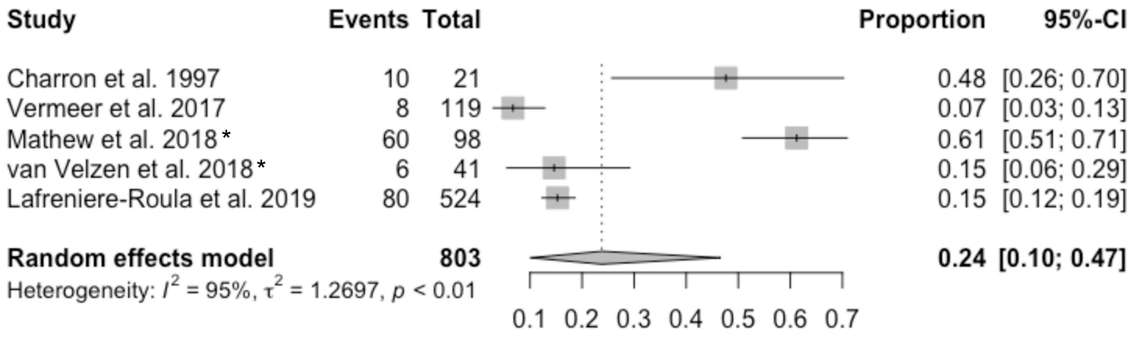
Supplementary Figure 12: Forest plot of penetrance across predominantly adult cohorts that both excluded and included probands.



Supplementary Figure 12 (cont.): Forest plot of penetrance across predominantly adult cohorts that both excluded and included probands.

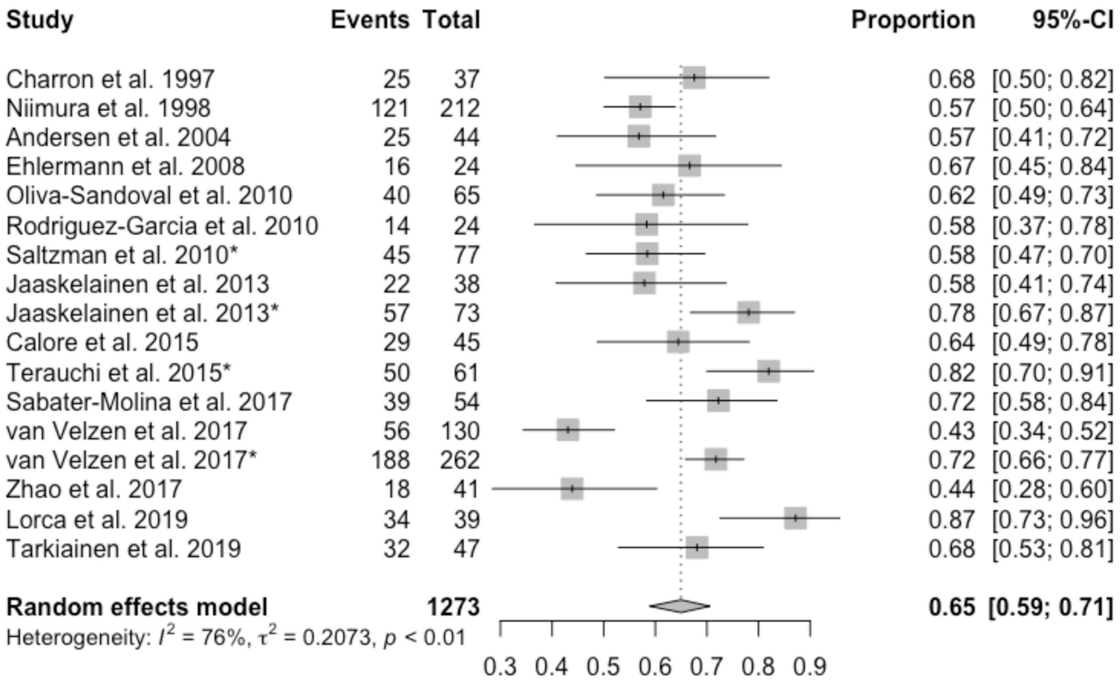


Supplementary Figure 13: Forest plot of penetrance before the age of 20 years across cohorts that both excluded and included probands. *Studies include probands.

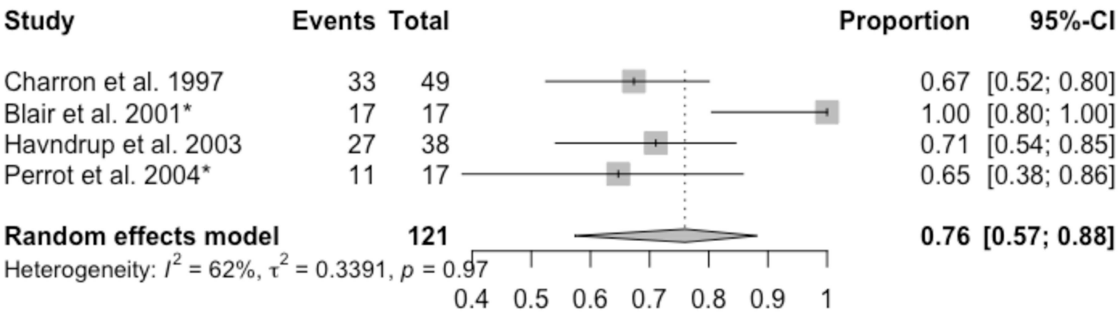


Supplementary Figure 14: Forest plot of (A) *MYBPC3*, (B) *MYH7*, and (C) *TNNT2* gene specific penetrance. *Studies include probands.

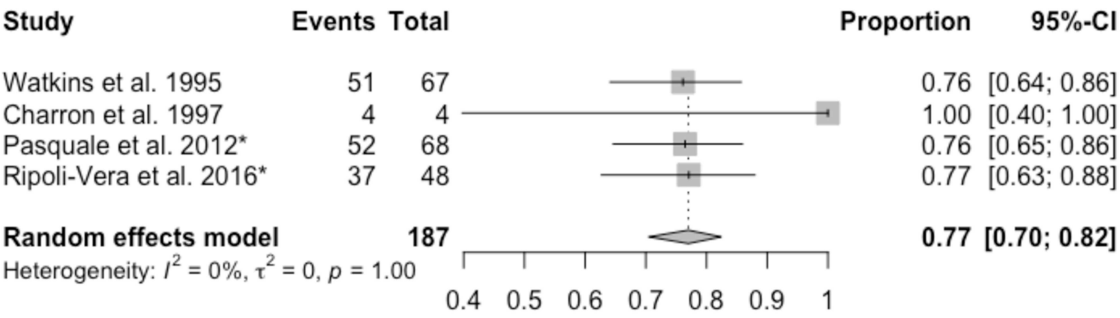
A. *MYBPC3*



B. *MYH7*



C. *TNNT2*



ST1 - All Included References (1 of 3)

Author	Year	PMID	DOI	Country	Detection Rate	Genotype/ Phenotype	Penetrance
Adalsteinsdottir et al	####	25078086	10.1161/CIRCULATIONAHA.114.011207	Iceland	X		
Agarwal et al.	####	26271555	10.1016/j.gheart.2015.01.001	United States	X	X	
Alfares et al.	####	25611685	10.1038/gim.2014.205	United States	X		
Andersen et al.	####	15114369	10.1038/sj.ejhg.5201190	Denmark	X		X
Andersen et al.	####	19035361	10.1002/humu.20862	Denmark	X		X
Arad et al.	####	16267253	10.1161/CIRCULATIONAHA.105.547448	United States			X
Bagnall et al.	####	30025578	10.1016/j.jacc.2018.04.078	Australia	X		
Bales et al.	####	26936621	10.1007/s00246-016-1358-y	United States			X
Biagini et al.	####	25037680	10.1016/j.amjcard.2014.05.065	Italy, United States	X	X	
Binder et al.	####	16610565	10.4065/81.4.459	United States	X	X	
Blair et al.	####	11861413	10.1161/hh0302.104532	United Kingdom, South Africa			X
Blauwet et al.	####	18835191	10.1016/j.carpath.2008.08.003	United States	X		
Bonaventura et al.	####	31110529	10.5114/aoms.2018.78767	Czech Republic	X	X	
Bongini et al.	####	26869393	10.1016/j.amjcard.2015.12.058	Italy, United States		X	
Bottillo et al.	####	26656175	10.1016/j.gene.2015.11.048	Italy, United States	X		
Brito et al.	####	22857948	10.1016/j.repc.2011.12.020	Portugal			X
Butters et al.	####	31930598	10.1111/jce.14346	Australia	X		
Calore et al.	####	25740977	10.1136/jmedgenet-2014-102923	Italy			X
Cardim et al.	####	29358015	10.1016/j.repc.2017.08.005	Portugal	X		
Charron et al.	####	9219008		France			X
Chida et al.	####	27885498	10.1007/s00380-016-0920-0	Japan		X	
Chiou et al.	####	25086479	10.1016/j.jjcc.2014.05.010	Taiwan	X	X	
Christiaans et al.	####	21459882	10.1093/eurheartj/ehr092	Netherlands			X
Chung et al.	####	31660989	10.1186/s12947-019-0171-1	South Korea	X		
Cirino et al.	####	29030401	10.1161/CIRCGENETICS.117.001768	United States	X		
Claes et al.	####	26497160	10.1093/eurheartj/ehv522	Netherlands			X
Curila et al.	####	22455086	10.1080/ac.67.1.2146562	Czech Republic		X	
Das et al.	####	24113344	10.1038/gim.2013.138	Australia	X		X
Doolan et al.	####	15698845	10.1016/j.yjmcc.2004.12.006	Australia			X
Ehlermann et al.	####	18957093	10.1186/1471-2350-9-95	Germany			X
Fokstuen et al.	####	21239446	10.1136/jmg.2010.083345	Germany, Italy, Switzerland	X		
Fourey et al.	####	28420666	10.1161/CIRCGENETICS.116.001685	Canada	X	X	
Frisso et al.	####	19659763	10.1111/j.1399-0004.2009.01190.x	Italy	X		
Fujita et al.	####	24621997	10.1016/j.jchf.2013.08.007	Japan	X	X	
Garcia-Pavia et al.	####	21896538	10.1093/eurjhf/hfr110	Spain	X		X
Gomez et al.	####	28356264	10.1161/CIRCGENETICS.116.001584	Spain		X	
Gray et al.	####	21862152	10.1016/j.ijcard.2011.07.095	Australia			X
Gruner et al.	####	21511876	10.1161/CIRCGENETICS.110.958835	Canada	X		
Gruner et al.	####	23239831	10.1161/CIRCGENETICS.112.963363	Canada	X	X	
Havndrup et al.	####	12566107	10.1016/s0008-6363(02)00711-3	Denmark			X
Hayashi et al.	####	29907873	10.1038/s10038-018-0479-y	Japan	X		
Ho et al.	####	30297972	10.1161/CIRCULATIONAHA.117.033200	NR	X	X	
Huurman et al.	####	31776912	10.1007/s12471-019-01349-1	NR			X
Ingles et al.	####	23598715	10.1038/gim.2013.44	Australia	X	X	
Ingles et al.	####	28408708	10.1161/CIRCGENETICS.116.001620	Australia	X		
Jaafar et al.	####	27574918	10.1089/gtmb.2016.0187	Spain, Tunisia	X		

ST1 - All Included References (2 of 3)

Author	Year	PMID	DOI	Country	Detection Rate	Genotype/ Phenotype	Penetrance
Jääskeläinen et al.	####	15000344	10.1080/07853890310017161	Finland	X		
Jääskeläinen et al.	####	22462493	10.3109/07853890.2012.671534	Finland			X
Jääskeläinen et al.	####	30775854	10.1002/ehf2.12420	NR	X	X	
Jalanko et al.	####	30497761	10.1016/j.jelectrocard.2018.07.009	Finland			X
Jensen et al.	####	23197161	10.1161/CIRCULATIONAHA.111.090514	Denmark			X
Kaski et al.	####	20031618	10.1161/CIRCGENETICS.108.821314	United Kingdom	X	X	
Kindel et al.	####	22555271	10.1016/j.cardfail.2012.01.017	United States	X		
Knight et al.	####	31229680	10.1016/j.hrthm.2019.06.015	United States			X
Ko et al.	####	28640247	10.1038/gim.2017.79	United States	X	X	
Lafreniere-Roula et al	####	31170284	10.1093/eurheartj/ehz396	Canada			X
Lee et al.	####	30354366	10.1161/CIRCHEARTFAILURE.118.005191	NR		X	
Li et al.	####	24909666	10.1161/CIRCGENETICS.113.000331	Canada	X	X	
Limongelli et al.	####	26808413	10.2459/JCM.0000000000000361	Italy	X		
Lipari et al.	####	31919335	10.20452/pamw.15130	Poland	X		
Loar et al.	####	26061417	10.1111/chd.12280	United States	X	X	
Lopes et al.	####	25351510	10.1136/heartjnl-2014-306387	United Kingdom	X	X	
Lopes et al.	####	30554928	10.1016/j.ijcard.2018.12.012	Portugal		X	
Lorca et al.	####	29631964	10.1016/j.rec.2018.02.009	Spain	X		X
Lu et al.	####	30165862	10.1186/s12967-018-1605-5	China	X		
Mademont-Soler et a	####	28771489	10.1371/journal.pone.0181465	Spain	X		
Mak et al.	####	30022097	10.1038/s41598-018-29263-3	China	X		
Maron et al.	####	21839045	10.1016/j.hrthm.2011.08.009	Australia, United States	X		
Mathew et al.	####	29053178	10.1111/cge.13157	Canada		X	X
Mattos et al.	####	27737317	10.5935/abc.20160133	Brazil			X
Maurizi et al.	####	29710196	10.1001/jamacardio.2018.0789	Italy	X	X	
Maurizi et al.	####	30715937	10.1161/CIRCULATIONAHA.118.037264	United States, Italy, Netherlands, Australia			X
Mazzarotto et al.	####	29875424	10.1038/s41436-018-0046-0	Italy	X		
McLeod et al.	####	19853701	10.1016/j.ahj.2009.09.006	United States	X		
Michels et al.	####	19666645	10.1093/eurheartj/ehp306	Netherlands			X
Miller et al.	####	23054336	10.1007/s10897-012-9544-4	United States	X		X
Miller et al.	####	31199839	10.1371/journal.pone.0217612	United States	X	X	
Mogensen et al.	####	15607392	10.1016/j.jacc.2004.05.088	United Kingdom			X
Monserrat et al.	####	17611253	10.1093/eurheartj/ehm239	Spain			X
Morita et al.	####	16754800	10.1161/CIRCULATIONAHA.105.593558	United States	X		
Morita et al.	####	18403758	10.1056/NEJMoa075463	United States	X	X	
Mörner et al.	####	12818575	10.1016/s0022-2828(03)00146-9	Sweden	X		
Murphy et al.	####	26914223	10.1007/s12265-016-9681-5	United States	X		
Naraen et al.	####	30348007	10.1177/2047487318808044	NR	X	X	
Neubauer et al.	####	31699273	10.1016/j.jacc.2019.08.1057	United States, Canada, United Kingdom, Italy, Germany, Netherlands	X	X	
Newman et al.	####	29362845	10.1007/s00246-018-1810-2	United States	X		
Nguyen et al.	####	31112424	10.1161/CIRCGEN.119.002500	France	X		
Niimura et al.	####	9562578	10.1056/NEJM199804303381802	Canada, United Kingdom, United States			X
Ntusi et al.	####	27841901	10.5830/CVJA-2015-075	South Africa	X	X	
Oliva-Sandoval et al.	####	21088121	10.1136/hrt.2010.200402	Spain			X
Olivotto et al.	####	18533079	10.4065/83.6.630	Italy	X		
Olivotto et al.	####	21835320	10.1016/j.jacc.2011.05.018	Italy, United States	X	X	

ST1 - All Included References (3 of 3)

Author	Year	PMID	DOI	Country	Detection Rate	Genotype/ Phenotype	Penetrance
Otsuka et al.	####	22112859	10.1253/circj.cj-11-0876	Japan	X		
Ouellette et al.	####	28369760	10.1111/cge.13024	Canada	X		
Pasquale et al.	####	22144547	10.1161/CIRCGENETICS.111.959973	United Kingdom			X
Perrot et al.	####	15856146	10.1007/s00109-005-0635-7	Germany, Kyrgyzstan, Poland			X
Alejandra Restrepo-C	####	28138913	10.1007/s12265-017-9730-8	Spain	X		X
Richard et al.	####	12707239	10.1161/01.CIR.0000066323.15244.54	France	X		
Ripoll-Vera et al.	####	26507537	10.1016/j.rec.2015.06.025	Spain			X
Robyns et al.	####	31513939	10.1016/j.ejmg.2019.103754	Belgium	X	X	
Rodríguez-García et a	####	20433692	10.1186/1471-2350-11-67	Spain			X
Ross et al.	####	28615295	10.1161/CIRCGENETICS.116.001671	Australia	X		
Rubattu et al.	####	27483260	10.3390/ijms17081239	Italy	X		
Rupp et al.	####	30105547	10.1007/s00392-018-1354-8	Germany	X		
Sabater-Molina et al.	####	28029522	10.1016/j.rec.2016.06.020	Spain			X
Saltzman et al.	####	20378854	10.1161/CIRCRESAHA.109.216291	United States			X
Seidelmann et al.	####	28087566	10.1161/CIRCGENETICS.116.001573	United States	X		
Song et al.	####	15563892	10.1016/j.cccn.2004.09.016	China		X	X
Tarkiainen et al.	####	30976029	10.1038/s41598-019-42376-7	Finland			X
Teramoto et al.	####	29398688	10.1253/circj.CJ-17-1012	Japan	X		
Terauchi et al.	####	25123604	10.1016/j.jjcc.2014.07.010	Japan			X
Tobita et al.	####	29386531	10.1038/s41598-018-20114-9	Japan	X	X	
Towe et al.	####	25495007	10.1111/chd.12242	United States	X	X	
Tran Vu et al.	####	31308319	10.1253/circj.CJ-19-0190	Vietnam	X	X	
Valdés-Mas et al.	####	25351925	10.1038/ncomms6326	Spain			X
Van Driest et al.	####	15519027	10.1016/j.jacc.2004.07.045	United States	X	X	
Van Driest et al.	####	15945527	10.1016/S0025-6196(11)61527-9	United States		X	
van Velzen et al.	####	27476098	10.1016/j.amjcard.2016.06.038	Netherlands	X	X	
van Velzen et al.	####	28794111	10.1161/CIRCGENETICS.116.001660	NR			X
van Velzen et al.	####	30292335	10.1016/j.amjcard.2018.08.040	Netherlands	X	X	
van Velzen et al.	####	29661763	10.1161/CIRCGEN.117.001896	Netherlands			X
Vermeer et al.	####	28396031	10.1016/j.jpeds.2017.03.033	Amsterdam, Netherlands			X
Vigneault et al.	####	30561279	10.1148/radiol.2018180339	NR			X
Viswanathan et al.	####	29121657	10.1371/journal.pone.0187948	United States			X
Waldmuller et al.	####	21750094	10.1093/eurjhf/hfr074	Germany		X	
Walsh et al.	####	28082330	10.1093/eurheartj/ehw603	Singapore, United Kingdom	X		
Wang et al.	####	18383048	10.1002/clc.20151	China		X	X
Wang et al.	####	25132132	10.1002/ejhf.144	China	X	X	
Watkins et al.	####	7898523	10.1056/NEJM199504203321603	United States			X
Weissler-Snir et al.	####	28193612	10.1161/CIRCIMAGING.116.005311	Canada, Italy, Switzerland, United States	X	X	
Zhao et al.	####	28450932	10.3892/etm.2017.4089	China			X
Zou et al.	####	23283745	10.1007/s11033-012-2474-2	China	X	X	
				TOTALS	80	44	51

LEGEND:

PMID = PubMed ID

ST2 - Detection Rate Adult (1 of 1)

			Detection Rate						Detection Rate by Family History												Mulitple Variant Cases						ACMG/AMP	≥8 Genes		
			Overall			Inconclusive			Sporadic			FHx HCM			FHx SCD			FHx CM +/- SCD			% of Positives			% of all Cases			Criteria	Tested in	# of Genes	
Author	Year	PMID	n	N	%	n	N	%	n	N	%	n	N	%	n	N	%	n	N	%	n	N	%	n	N	%	Applied	Full Cohort	Tested	Quality Assessments
Mörner et al.	2003	12818575	13	46	28.3%	.	.	.	3	35	8.6%	10	11	90.9%	.	.	.	10	11	90.9%	.	.	.	.	.	.	N	Y	8	Low/Poor
Richard et al.	2003	12707239	124	197	62.9%	.	.	.	15	25	60.0%	109	172	63.4%	.	.	.	109	172	63.4%	.	.	.	.	.	.	N	N	6-9	Low/Poor
Andersen et al.	2004	15114369	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	10	81	12.3%	.	.	.	.	.	.	N	Y	8	Low/Poor
Jääskeläinen et al.	2004	15000344	14	35	40.0%	.	.	.	.	.	.	14	23	60.9%	.	.	.	14	23	60.9%	.	.	.	.	.	.	N	Y	9	Low/Poor
Van Driest et al.	2004	15519027	147	389	37.8%	.	.	.	.	.	.	65	120	54.2%	29	56	51.8%	65	120	54.2%	.	.	.	.	.	.	N	Y	8	Low/Poor
Binder et al.	2006	16610565	143	382	37.4%	.	.	.	.	.	.	67	124	54.0%	36	75	48.0%	67	124	54.0%	.	.	.	.	.	.	N	Y	8	Low/Poor
Morita et al.	2006	16754800	8	50	16.0%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	11	Low/Poor
Andersen et al.	2009	19035361	29	85	34.1%	.	.	.	5	11	45.5%	.	.	.	.	.	.	23	43	53.5%	.	.	.	.	.	.	N	Y	11	Low/Poor
Olivotto et al.	2008	18533079	126	203	62.1%	.	.	.	.	.	.	44	57	77.2%	23	29	79.3%	44	57	77.2%	.	.	.	.	.	.	N	Y	8	Low/Poor
Blauwet et al.	2009	18835191	18	59	30.5%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	8	Low/Poor
McLeod et al.	2009	19853701	76	181	42.0%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	8	Low/Poor
Fokstuen et al.	2011	21239446	38	122	31.1%	.	.	.	.	.	.	.	.	.	.	.	.	28	62	45.2%	.	.	.	.	.	.	N	Y	12	Very low/very poor
Garcia-Pavia et al.	2011	21896538	12	24	50.0%	.	.	.	.	.	.	9	15	60.0%	5	8	62.5%	9	15	60.0%	.	.	.	.	.	.	N	Y	10	Low/Poor
Gruner et al.	2011	21511876	153	425	36.0%	.	.	.	.	.	.	4	10	40.0%	1	4	25.0%	4	10	40.0%	.	.	.	.	.	.	N	Y	11	Low/Poor
Olivotto et al.	2011	21835320	42	61	68.9%	.	.	.	.	.	.	17	25	68.0%	5	9	55.6%	17	25	68.0%	.	.	.	.	.	.	N	Y	8	Low/Poor
Otsuka et al.	2012	22112859	.	.	.	.	.	.	.	.	.	49	112	43.8%	.	.	.	49	112	43.8%	.	.	.	.	.	.	N	Y	8	Low/Poor
Fujita et al.	2013	24621997	90	193	46.6%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	9	Low/Poor
Gruner et al.	2013	23239831	163	471	34.6%	70	471	14.9%	.	.	.	88	145	60.7%	28	51	54.9%	88	145	60.7%	.	.	.	.	.	.	N	Y	11	Low/Poor
Ingles et al.	2013	23598715	138	265	52.1%	6	265	2.3%	36	124	29.0%	102	141	72.3%	57	65	87.7%	102	141	72.3%	.	.	.	.	.	.	N	Y	8-10	Low/Poor
Miller et al.	2013	23054336	34	46	73.9%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	34	Low/Poor
Zou et al.	2013	23283745	102	200	51.0%	.	.	.	44	105	41.9%	.	.	.	.	.	.	58	95	61.1%	.	.	.	.	.	.	N	Y	8	Low/Poor
Adalsteinsdottir et al.	2014	25078086	101	151	66.9%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	N	up to 11	Low/Poor
Biagini et al.	2014	25037680	104	156	66.7%	.	.	.	.	.	.	71	97	73.2%	41	57	71.9%	71	97	73.2%	.	.	.	.	.	.	N	Y	10	Low/Poor
Chiou et al.	2015	25086479	13	38	34.2%	.	.	.	.	.	.	.	.	.	5	7	71.4%	5	7	71.4%	.	.	.	.	.	.	N	Y	6-8	Low/Poor
Das et al.	2014	24113344	63	136	46.3%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	N	7-10+	Low/Poor
Li et al.	2014	24909666	198	558	35.5%	.	.	.	.	.	.	104	176	59.1%	33	59	55.9%	104	176	59.1%	12	198	6.1%	12	558	2.2%	N	Y	11	Low/Poor
Wang et al.	2014	25132132	232	529	43.9%	.	.	.	.	.	.	83	130	63.8%	47	77	61.0%	83	130	63.8%	.	.	.	.	.	.	N	Y	8	Low/Poor
Agarwal et al.	2015	26271555	57	124	46.0%	.	.	.	.	.	.	22	31	71.0%	.	.	.	22	31	71.0%	.	.	.	.	.	.	N	Y	18	Low/Poor
Alfares et al.	2015	25611685	778	2412	32.3%	349	2412	14.5%	.	.	.	491	938	52.3%	134	402	33.3%	491	938	52.3%	11	778	1.4%	11	2412	0.5%	N	N	5-51	Low/Poor
Bottillo et al.	2016	26656175	33	41	80.5%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	62	Low/Poor
Loar et al.	2015	26061417	284	916	31.0%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	9	Low/Poor
Lopes et al.	2015	25351510	383	874	43.8%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	41	Low/Poor
Towe et al.	2015	25495007	18	71	25.4%	.	.	.	.	.	.	7	21	33.3%	4	17	23.5%	7	21	33.3%	.	.	.	.	.	.	N	Y	9	Low/Poor
Jaafar et al.	2016	27574918	12	45	26.7%	.	.	.	.	.	.	3	11	27.3%	5	15	33.3%	5	15	33.3%	.	.	.	.	.	.	N	Y	10	Low/Poor
Murphy et al.	2016	26914223	101	198	51.0%	.	.	.	.	.	.	55	72	76.4%	40	64	62.5%	55	72	76.4%	.	.	.	.	.	.	N	N	1-31	Low/Poor
Ntusi et al.	2016	27841901	10	35	28.6%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	15	Low/Poor
van Velzen et al.	2016	27476098	234	512	45.7%	.	.	.	.	.	.	.	.	.	46	61	75.4%	46	61	75.4%	.	.	.	.	.	.	N	Y	11-48	Low/Poor
Alejandra Restrepo-Cordoba et al.	2017	28138913	51	100	51.0%	.	.	.	.	.	.	24	42	57.1%	12	31	38.7%	24	42	57.1%	.	.	.	.	.	.	N	Y	10	Low/Poor
Fourey et al.	2017	28420666	272	1411	19.3%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	272	0.4%	1	1411	0.1%	Y	N	5-51	Low/Poor
Ingles et al.	2017	28408708	190	413	46.0%	21	413	5.1%	74	251	29.5%	116	162	71.6%	.	.	.	116	162	71.6%	.	.	.	.	.	.	N	N	7-46	Good quality
Ko et al.	2017	28640247	56	120	46.7%	11	120	9.2%	.	.	.	34	41	82.9%	.	.	.	34	41	82.9%	.	.	.	.	.	.	N	Y	8+	Low/Poor
Limongelli et al.	2017	26808413	52	94	55.3%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	10	Low/Poor
Mademont-Soler et al.	2017	28771489	111	303	36.6%	109	303	36.0%	.	.	.	.	.	.	.	.	.	.	.	.	6	129	4.7%	6	387	1.6%	Y	Y	5-25	Low/Poor
Ross et al.	2017	28615295	185	467	39.6%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	8	Low/Poor
Walsh et al.	2017	28082330	254	781	32.5%	.	.	.	.	.	.	95	147	64.6%	73	141	51.8%	95	147	64.6%	.	.	.	.	.	.	N	Y	38+	Low/Poor
Weissler-Snir et al.	2017	28193612	159	358	44.4%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	Y	Y	10-11	Low/Poor
Cardim et al.	2018	29358015	210	528	39.8%	40	528	7.6%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	N	up to 9+	Very low/very poor
Ho et al.	2018	30297972	998	2405	41.5%	229	2405	9.5%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	N	up to 8+	Low/Poor
Lu et al.	2018	30165862	12	34	35.3%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	Y	Y	64	Very low/very poor
Naraen et al.	2020	30348007	44	126	34.9%	.	.	.	.	.	.	.	.	.	.	.	.	26	47	55.3%	.	.	.	.	.	.	N	Y	19	Low/Poor
Teramoto et al.	2018	29398688	44	82	53.7%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	21-174	Low/Poor
Tobita et al.	2018	29386531	24	52	46.2%	11	52	21.2%	.	.	.	.	.	.	9	18	50.0%	16	37	43.2%	.	.	.	.	.	.	N	Y	95	Low/Poor
van Velzen et al.	2018	30292335	430	810	53.1%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	16	430	3.7%	16	810	2.0%	N	N	8-52	Low/Poor
Bonaventura et al.	2019	31110529	70	336	20.8%	72	336	21.4%	.	.	.	15	70	21.4%	6	16	37.5%	15	70	21.4%	1	70	1.4%	1	336	0.3%	Y	Y	229	Low/Poor
Chung et al.	2019	31660989	68	211	32.2%	.	.	.	.	.	.	.	.	.	6	14	42.9%	6	14	42.9%	.	.	.	.	.	.	Y	Y	82	Low/Poor
Jääskeläinen et al.	2019	30775854	146	382	38.2%	58	382	15.2%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	59	Low/Poor
Lorca et al.	2019	29631964	132	384	34.4%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	Y	Y	up to 10	Very low/very poor
Mazzarotto et al.	2019	29875424	491	1052	46.7%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	Y*	Y	3-48	Low/Poor
Miller et al.	2019	31199839	90	202	44.6%	13	202	6.4%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	8+	Low/Poor
Neubauer et al.	2019	31699273	774	2661	29.1%	325	2661	12.2%	.	.	.	.	.	.	.	.	.	.	.	.	3	774	0.4%	3	2661	0.1%	N	Y	36	Low/Poor
Robyns et al.	2020	31513939	141	378	37.3%	63	378	16.7%	.	.	.	79	134	59.0%	46	77	59.7%	79	134	59.0%	.	.	.	.	.	.	Y	N	3-34	Low/Poor
Tran Vu et al.	2019	31308319	44	104	42.3%	28	104	26.9%	.	.	.	13	19	68.4%	10	11	90.9%	13	19	68.4%	.	.	.	.	.	.	Y	Y	23	Low/Poor
Butters et al.	2020	31930598	271	844	32.1%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	Y	Y	8	Low/Poor
Lipari et al.	2020	31919335	12	29	41.4%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	1	12	8.3%	1	29	3.4%	Y	Y	404	Low/Poor

Legend

PMID = PubMed ID
FHx = family history
HCM = hypertrophic cardiomyopathy
SCD = sudden cardiac death
CM = Cardiomyopathy
ACMG = American College of Medical Genetics
AMP = Association of Molecular Pathology
. = not applicable
N = No
Y = Yes
* = ACMG/AMP citeria partially applied

ST3 - Detection Rates Peds (1 of 1)

			Detection Rate						Detection Rate by Family History												Mulitple Variant Cases						ACMG/AMP	≥8 Genes		
			Overall			Inconclusive			Sporadic			FHx HCM			FHx SCD			FHx CM +/- SCD			% of Positives			% of all Cases			Criteria	Tested in	# of Genes	
Author	Year	PMID	n	N	%	n	N	%	n	N	%	n	N	%	n	N	%	n	N	%	n	N	%	n	N	%	Applied	Full Cohort	Tested	Quality Assessments
Andersen et al.	2009	19035361	3	5	60.0%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	11	Low/Poor
Morita et al.	2008	18403758	46	84	54.8%	.	.	.	25	51	49.0%	21	33	63.6%	.	.	.	21	33	63.6%	.	.	.	.	.	.	N	Y	10	Low/Poor
Frisso et al.	2009	19659763	16	39	41.0%	.	.	.	4	18	22.2%	12	21	57.1%	.	.	.	12	21	57.1%	.	.	.	.	.	.	N	Y	8	Low/Poor
Kaski et al.	2009	20031618	42	79	53.2%	.	.	.	14	31	45.2%	28	48	58.3%	15	28	53.6%	28	48	58.3%	.	.	.	.	.	.	N	Y	11	Low/Poor
Kindel et al.	2012	22555271	29	42	69.0%	.	.	.	.	.	.	14	33	42.4%	6	14	42.9%	14	33	42.4%	.	.	.	.	.	.	N	N	5 - 17	Low/Poor
Alfares et al.	2015	25611685	129	462	27.9%	87	462	18.8%	.	.	.	.	.	.	.	.	.	.	.	.	6	129	4.7%	6	462	1.3%	N	N	5 - 51	Low/Poor
Loar et al.	2015	26061417	71	137	51.8%	.	.	.	.	.	.	.	.	.	.	.	.	39	78	50.0%	.	.	.	.	.	.	N	Y	9	Low/Poor
Ouellette et al.	2017	28369760	25	66	37.9%	21	66	31.8%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	Y*	N	5 - 62	Low/Poor
Hayashi et al.	2018	29907873	36	46	78.3%	.	.	.	20	26	76.9%	.	.	.	.	.	.	16	20	80.0%	.	.	.	.	.	.	Y	Y	67	Low/Poor
Newman et al.	2018	29362845	32	77	41.6%	.	.	.	.	.	.	14	19	73.7%	10	23	43.5%	14	23	60.9%	2	32	6.3%	2	77	2.6%	N	N	5-36	Low/Poor
Maurizi et al.	2018	29710196	55	70	78.6%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	Y	Y	8-12	Low/Poor
Rupp et al.	2019	30105547	29	36	80.6%	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	N	Y	28	Low/Poor

Legend
PMID = PubMed ID
FHx = family history
HCM = hypertrophic cardiomyopathy
SCD = sudden cardiac death
CM = Cardiomyopathy
ACMG = American College of Medical Genetics
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ST4 - Exomes Genomes (1 of 1)

Author	Year	PMID	Exome/Genome Sequencing Detection Rate	Comments	Quality Assessments
Seidelmann et al.	2017	28087566	*Exome:* 13/28 (46%) Proband only analyses	10/13 results were in one of the 8 definitively established sarcomere HCM genes	Low/Poor
Cirino et al.	2017	29030401	*Genome:* 20/41 (49%) 19/20 variants previously identified by panel testing were identified 1/20 variants found in previously negative case		Low/Poor
Bagnall et al.	2018	30025578	*Genome:* Cohort 1 - Probands with previously negative genetic testing 9/46 (20%) Proband only (n=33), Duos (n=10), Trios (n=3) Cohort 2 - Probands with no prior testing 4/12 (33%) Proband only	Cohort 1: 4 results in genes not previously tested 4 results due to deep intronic variants Cohort 2: 3 results due to variants in common HCM genes (*MYBPC3* , *MYH7* , and *TNNT2*)	Low/Poor
Mak et al.	2018	30022097	*Exome:* 6/14 (43%) Specific analysis type unspecified (e.g., proband only, duos, or trios)	Noted exome coverage of TNNI3 was generally poor	Low/Poor
Nguyen et al.	2019	31112424	*Exome:* 175/200 (88%) Specific analysis type unspecified (e.g., proband only, duos, or trios)	Majority of results were in genes other than the 8 definitively established sarcomere HCM genes Limited details on the variant classification approach	Very low/very poor

Legend
PMID = PubMed ID

ST5 - Detection Rate by Genes (1 of 1)

			CASES							VARIANTS										
Author	Year	PMID	N	MYBPC3 + MYH7	%	Def. Assoc. HCM Genes	%	Other Genes	%	N	MYBPC3 + MYH7	%	Def. Assoc. HCM Genes	%	Other Genes	%	ACMG/AMP Criteria Applied	≥8 Genes Tested in Full Cohort	# of Genes Tested	Quality Assessments
Adults																				
Gruner et al.	2013	23239831	163	134	82.2%	158	96.9%	5	3.1%	.	.	.	.	.	.	.	N	Y	11	Low/Poor
Das et al.	2014	24113344	.	.	.	.	.	.	.	72	57	79.2%	66	91.7%	6	8.3%	N	N	7-10+	Low/Poor
Li et al.	2014	24909666	.	.	.	.	.	.	.	193	162	83.9%	193	100.0%	0	0.0%	N	Y	11	Low/Poor
Alfares et al.	2015	25611685	923	767	83.1%	898	97.3%	25	2.7%	.	.	.	.	.	.	.	N	N	5-51	Low/Poor
Murphy et al.	2016	26914223	101	83	82.2%	93	92.1%	8	7.9%	.	.	.	.	.	.	.	N	N	1-31	Low/Poor
Ingles et al.	2017	28408708	.	.	.	.	.	.	.	190	163	85.8%	190	100.0%	0	0.0%	N	N	7-46	Good quality
Ross et al.	2017	28615295	185	165	89.2%	185	100.0%	0	0.0%	.	.	.	.	.	.	.	N	Y	8	Low/Poor
Weissler-Snir et al.	2017	28193612	159	128	80.5%	157	98.7%	2	1.3%	.	.	.	.	.	.	.	Y	Y	10-11	Low/Poor
Cardim et al.	2018	29358015	210	155	73.8%	201	95.7%	8	3.8%	.	.	.	.	.	.	.	N	N	up to 9+	Very low/very poor
Ho et al.	2018	30297972	.	.	.	.	.	.	.	707	594	84.0%	662	93.6%	45	6.4%	N	N	up to 8+	Low/Poor
Lu et al.	2018	30165862	12	10	83.3%	10	83.3%	2	16.7%	.	.	.	.	.	.	.	Y	Y	64	Very low/very poor
Naraen et al.	2020	30348007	44	33	75.0%	40	90.9%	4	9.1%	.	.	.	.	.	.	.	N	Y	19	Low/Poor
Bonaventura et al.	2019	31110529	.	.	.	.	.	.	.	72	56	77.8%	65	90.3%	7	9.7%	Y	Y	229	Low/Poor
Chung et al.	2019	31660989	.	.	.	.	.	.	.	71	52	73.2%	67	94.4%	4	5.6%	Y	Y	82	Low/Poor
Mazzarotto et al.	2019	29875424	.	.	.	.	.	.	.	524	437	83.4%	524	100.0%	0	0.0%	Y*	Y	3-48	Low/Poor
Miller et al.	2019	31199839	90	78	86.7%	.	.	.	.	.	.	.	.	.	.	.	N	Y	8+	Low/Poor
Robyns et al.	2020	31513939	141	107	75.9%	124	87.9%	17	12.1%	.	.	.	.	.	.	.	Y	N	3-34	Low/Poor
Tran Vu et al.	2019	31308319	44	25	56.8%	41	93.2%	3	6.8%	.	.	.	.	.	.	.	Y	Y	23	Low/Poor
Butters et al.	2020	31930598	271	232	85.6%	.	.	.	.	.	.	.	.	.	.	.	Y	Y	8	Low/Poor
Lipari et al.	2020	31919335	12	10	83.3%	11	91.7%	1	8.3%	.	.	.	.	.	.	.	Y	Y	404	Low/Poor
Pediatrics																				
Kindel et al.	2012	22555271	29	17	58.6%	18	62.1%	11	37.9%	.	.	.	.	.	.	.	N	N	5 to 17	Low/Poor
Ouellette et al.	2017	28369760	25	24	96.0%	.	.	.	.	.	.	.	.	.	.	.	Y*	N	5 to 62	Low/Poor
Hayashi et al.	2018	29907873	.	.	.	.	.	.	.	42	28	66.7%	34	81.0%	8	19.0%	Y	Y	67	Low/Poor
Newman et al.	2018	29362845	32	25	78.1%	31	96.9%	1	3.1%	.	.	.	.	.	.	.	N	N	5-36	Low/Poor

Legend

PMID = PubMed ID

Def. Assoc. = Definitively associated

HCM = hypertrophic cardiomyopathy

ACMG = American College of Medical Genetics

AMP = Association of Molecular Pathology

. = not applicable

N = No

Y = Yes

* = ACMG/AMP citeria partially applied

ST6 - Geno Pheno Age of Onset (1 of 3)

Genotype+ v. Genotype-				Mean age of onset			Median age of onset			
Author	Year	PMID	N	*G+*	*G-*	p-value	*G+*	*G-*	p-value	Quality Assessment
Adult/Mixed Peds and Adult										
Agarwal et al.	2015	26271555	G+=57 G-=67	50+/-17.9	59.1+/-13.9	p=0.01	.	.	.	fair
Li et al.	2014	24909666	G+=198 G-=360	39.5+/-15.2	48.5+/-14.8	p<0.001	.	.	.	low/poor
Towe et al.	2015	25495007	G+=18 G-=53	44.2+/-13.5	49 +/-15.7	p=0.4	.	.	.	very low/very poor
Ingles et al.	2013	23598715	G+=138 G-=127	47+/-16	56+/-17	p<0.001	.	.	.	low/poor
Binder et al.	2006	16610565	G+=143 G-=239	35.7+/-17	45.1+/-19	p=<.05	.	.	.	fair
Chiou et al.	2015	25086479	G+=13 G-=25	47.1+/-14.0	67.2+/-12.2	p=<.05	.	.	.	very low/very poor
Gomez et al.	2017	28356264	G+=169 G-=279	44+/-16	51+/-17	p<0.0001	.	.	.	very low/very poor
Gruner et al.	2013	23239831	G+=163 G-=308	38.5+/-16.2	48.7+/-14.6	p<0.001	.	.	.	low/poor
Ko et al.	2017	28640247	G+=67 G-=54	38+/15	54+/-13	p<0.001	.	.	.	very low/very poor
Jääskeläinen et a	2019	30775854	G+=255 G-=227	47.8 +/-14.6	54.1 +/- 14.8	p=<0.001	.	.	.	fair
Lopes et al.	2015	25351510	G+=383 G-=491	45.8+/-14.7	53.1 +/- 14.9	p=2.23e^-11	.	.	.	good
Van Driest et al.	2005	15945527	G+=147 G-=242	35.6	44.9	p=0.000	.	.	.	very low/very poor
Olivotto et al.	2011	21835320	G+=126 G-=77	34+/-13	31+/-14	p=0.41	.	.	.	low/poor
Tran Vu et al.	2019	31308319	G+=44 G-=60	42.7+/-15.2	52.7+/-15	p<0.05	.	.	.	low/poor
Tobita et al.	2018	29386531	G+=24 G-=17	34.3+/-17.4	31.9+/-17.1	p=0.662	.	.	.	low/poor
Naraen et al.	2020	30348007	G+=44 G-=82	.	.	.	50.2 (34.1-61.8)	58.6 (50.0-66.5)	p= 0.007	very low/very poor
Lopes et al.	2019	30554928	G+=122 G-=301	46+/-18	53+/-16	p=<0.001	.	.	.	poor
Bonaventura et :	2019	31110529	G+=70 G-=266	44.5 +/- 15.6	55.6 +/- 13.9	p=<0.01	.	.	.	low/poor
Pediatric										
Loar et al.	2015	26061417	G+=71 G-=66	13.4+/-6	13.2+/-7	NS	.	.	.	low/poor
Kaski et al.	2009	20031618	G+=42 G-=37	.	.	.	9.4 (2.5-11)	7 (0.8-11)	p=0.731	very low/very poor
Morita et al.	2008	18403758	G+=46 G-=38	Sporadic: 6.5+/-6.6 Familial: 10.3+/-5.0	Sporadic: 5.3+/-5.6 Familial: 10.4+-6.7	Sporadic : p=0.47 Familial: p=0.96	.	.	.	low/poor

ST6 - Geno Pheno Age of Onset (2 of 3)

MYBPC3 v.				Mean age of onset			Median age of onset			
Author	**Year**	**PMID**	**N**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult										
Biagini et al.	2014	25037680	MYBPC3=38 MYH7=26	.	.	.	41 (25-55)	16 (10-30)	NR	fair
Bongini et al.	2016	26869393	MYBPC3=137 MYH7=67	41+/-15	38+/-18	p=0.22	.	.	.	low/poor
Chida et al.	2017	27885498	MYBPC3=19 MYH7=31	.	.	.	24 (12.7-51.1)	5 (11.6-25.5)	NR	very low/very poor
Curila et al.	2012	22455086	MYBPC3=21 MYH7=11	34+/-14	28+/-18	p=0.3051	.	.	.	low/poor
Ingles et al.	2013	23598715	MYBPC3=20 MYH7=44	40 +/- 17	27 +/- 15	p<0.001	.	.	.	low/poor
Song et al.	2005	15563892	MYBPC3=16 MYH7=52	48.4+-13.5	34.6+-14.0	NR	.	.	.	very low/very poor
Waldmuller et al	2011	21750094	MYBPC3=52 MYH7-35	35.4+-15.3	34.4+-19.0	NS	.	.	.	low/poor
Wang et al.	2008	18383048	MYBPC3=18 MYH7=52	50.0 ± 15	34.6 ± 14.0	p<0.001	.	.	.	low/poor
Tobita et al.	2018	29386531	MYBPC3=11 MYH7=12	39.0+/-12.5	25.4+/-17.6	NS	.	.	.	low/poor
Robyns et al.	2020	31513939	MYBPC3=80 MYH7=274	37.76+/-16.56	34.00+/-19.79	p<0.001				low/poor
Lopes et al.	2019	30554928	MYBPC3=62 MYH7=37	49+/-17	41+/-17	NR	.	.	.	poor
Lee et al.	2018	30354366	MYBPC3=659 MYH7=296	41.8+/-14.7	37.0+/-15.5	p=0.049	.	.	.	good
Pediatric										
Loar et al.	2015	26061417	MYBPC3=24 MYH7=29	NR	NR	NS	.	.	.	low/poor
Maurizi et al.	2018	29710196	MYBPC3=20 MYH7=19	Diagnosed<12yo: OR=1.25 (0.49-3.17) Diagnosed>12yo: OR=1.33 0.56-3.16)	.	NS	.	.	.	fair

ST6 - Geno Pheno Age of Onset (3 of 3)

***MYBPC3* v.**				**Mean age of onset**			**Median age of onset**			
Author	**Year**	**PMID**	**N**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult										
Van Driest et al.	2004	15519027	Multiple=10 Single=137	20.7	36.1	p=0.004	.	.	.	low/poor
Wang et al.	2014	25132132	Multiple=38 Single=194	39.1 +/-14.9	43.7 +/-15.5	p<10(-6)	.	.	.	fair
Zou et al.	2013	23283745	Multiple=19 Single=83	34.5 ± 12.9	38.4 ± 12.7	p=<0.01	.	.	.	low/poor
Biagini et al.	2014	25037680	Multiple=21 Single=83	.	.	.	28 (17-33)	30 (16-47)	NR	fair
Ingles et al.	2013	23598715	Multiple=16 Single=122	31+/- 17	35 +/- 17	p=0.38	.	.	.	low/poor
Fourey et al.	2017	28420666	Multiple=24 Single=140	33+/-15	40+/-14	p=0.09	.	.	.	low/poor
Pediatric										
Loar et al.	2015	26061417	Multiple=8 Single=63	NR	NR	NS	.	.	.	low/poor
Mathew et al.	2018	29053178	Multiple=16 Single=82	.	.	.	8.9(3.1-11.2)	13.1 (6.9, 15.8)	p=0.014	fair

LEGEND

PMID = PubMed ID
G+ = genotype postiive cohort
G- = genotype negative cohort
NS = not significant
NR = not reported (when it could have been reported)
. = not applicable

ST7- Geno Pheno SCA (1 of 2)

Genotype+ v. Genotype-				SCD only			SCA only			ICD tx only			Combination			
Author	**Year**	**PMID**	**N**	***G+***	***G-***	***p-value***	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																
Lopes et al.	2015	25351510	G+=383 G-=491	.	.	.	.	.	.	.	.	.	SCD/ICD tx Time from first eval.: HR 2.89 (1.01-8.33) Time from birth: 3.44 (1.19-9.92)		Time from first eval: p=0.039 Time from birth: p=0.028	Good
Lopes et al.	2019	30554928	G+=122 G-=302	.	.	.	.	.	.	.	.	.	SCD/SCA/ICDtx: G+ associated with SCD HR NR		p=0.017	Poor
van Velzen et al.	2018	30292335	G+=430 G-=280	.	.	.	.	.	.	.	.	.	SCD/SCA/ICDtx: HR 1.03 (0.56-1.90)	p=0.9	p=0.93	Low/Poor
Li et al.	2014	24909666	G+=198 G-=360	.	.	.	.	.	.	.	.	.	9/198 SCD/SCA/ICDtx: HR 1.35 (0.56-3.28)	11/360	p=0.349 p=0.50	Low/Poor
Towe et al.	2015	25495007	G+=18 G-=53	0/17	1/51	p=1.0	.	.	.	0/17	2/51	p=1.0	.	.	.	Very Low/Very Poor
van Velzen et al.	2016	27476098	G+=327 G-=299	7/327	2/299	.	.	.	.	.	.	.	SCA/ICDtx: 12/327 SCD/SCA/ICDtx: 19/327 SCD/SCA/ICDtx: HR=2.88(1.24-6.67)	SCA/ICDtx: 8/299 SCD/SCA/ICDtx:10/299 p=0.014		
Ingles et al.	2013	23598715	G+=138 G-=127	.	.	.	.	.	.	.	.	.	SCD/SCA: 14/138	SCD/SCA: 12/127	NS	Low/Poor
Tobita et al.	2018	29386531	G+=24 G-=17	.	.	.	3/24	3/17	p=0.679	.	.	.	.	.	.	Low/Poor
Pediatric																
Morita et al.	2008	18403758	G+=46 G-=38	3/46	0/38	NR	.	.	.	.	.	.	.	.	.	Low/Poor

ST7- Geno Pheno SCA (2 of 2)

MYBPC3 v. MYH7				SCD only			SCA only			ICD tx only			Combination			
Author	**Year**	**PMID**	**N**	***MYBPC3***	***MYH7***	***p-value***	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																
Bongini et al.	2016	26869393	MYBPC3+=137 MYH7+=67	.	.	.	5/137	1/67	NR	.	.	.	SCD/ICDtx: 5/137	SCD/ICDtx: 2/67	NR	Low/Poor
Ingles et al.	2013	23598715	MYBPC3+=20 MYH7+=44	.	.	.	.	.	.	.	.	.	SCD/SCA: 5/53	SCD/SCA: 4/44	NR	Low/Poor
Wang et al.	2008	18383048	MYBPC3+=18 MYH7+=52	0/18	7/52	p<0.001	.	.	.	.	.	.	.	.	.	Low/Poor
Weissler-Snir et al.	2017	28193612	MYBPC3+=75 MYH7+=53	1/75	2/53	NS	.	.	.	.	.	.	.	.	.	Low/Poor
Tobita et al.	2018	29386531	MYBPC3+=11 MYH7+=12	2/11	3/12	p=1	.	.	.	.	.	.	.	.	.	Low/Poor
Miller et al.	2019	31199839	MYBPC3+=37 MYH7+=41	.	.	.	.	.	.	4/20	2/11	p=1.00	.	.	.	Good
Ho et al.	2018	30297972	MYBPC3+=745 MYH7+=377		HR 0.59 (0.20-1.75)	NR		HR 0.34 (0.12-0.97)	NR		HR 0.81 (0.43-1.53)	NR		SCD/SCA/ICDtx: HR 0.68 (0.41-1.12)	NR	Good

Multiple v. Single				SCD only			SCA only			ICD tx only			Combination			
Author	**Year**	**PMID**	**N**	***Multiple***	***Single***	***p-value***	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																
Wang et al.	2014	25132132	Multiple=38 Single=194	5/38 HR 4.04 (1.27-12.86)	7/194	NR p=0.018	.	.	.	.	.	.	.	.	.	Fair
Ingles et al.	2013	23598715	Multiple=16 Single=122	.	.	.	.	.	.	.	.	.	SCD/SCA: 5/16	SCD/SCA: 9/122	p<0.01	Low/Poor
Fourey et al.	2017	28420666	Multiple=24 Single=140	.	.	.	.	.	.	.	.	.	SCD/SCA/ICDtx: 2/24	SCD/SCA/ICDtx: 7/140	p=0.62	Low/Poor
Ho et al.	2018	30297972	Multiple=34 Single=1,210	.	.	.	HR 2.3 (0.55-9.58)		NR	HR 5.75 (0.55-60.31)		NR	SCD/SCA/ICDtx: HR 1.42 (0.52-3.88)		NR	Good
Pediatric																
Mathew et al.	2018	29053178	Multiple=16 Single=82	.	.	.	HR 10.25 (0.93-113.3)		p=0.06	.	.	.	.	.	.	Fair

LEGEND

PMID = PubMed ID
G+ = genotype postiive cohort
G- = genotype negative cohort
SCD = sudden cardiac death
SCA = sudden cardiac arrest
ICD tx = implantable cardioverter defibrillator shock
NS = not significant
NR = not reported (when it could have been reported)
HR = hazard ratio
. = not applicable

ST8 - Geno Pheno ICD (1 of 2)

Genotype+ v. Genotype-				ICD			
Author	**Year**	**PMID**	**N**	***G+***	***G-***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult							
Van Driest et al.	2004	15519027	G+=147 G-=242	37/147	23/242	p=0.0001	low/poor
Agarwal et al.	2015	26271555	G+=57 G-=67	20/57	21/67	p=0.64	fair
Fujita et al.	2013	24621997	G+=78 G-=98	9/78	10/98	p>0.05	fair
Li et al.	2014	24909666	G+=198 G-=360	49/198	58/360	p=0.24	low/poor
Lopes et al.	2015	25351510	G+=383 G-=491	96/383	80/491	p=0.00128	good
Towe et al.	2015	25495007	G+=18 G-=53	6/17	15/51	p=0.80	very low/very poor
van Velzen et al.	2016	27476098	G+=234 G-=278	Primary prevention ICD: 8/234 Secondary prevention ICD: 11/234	Primary prevention ICD: 26/278 Secondary prevetion ICD: 9/278	Primary prevention ICD: p=0.019 Secondary prevention ICD: p=0.395	good
Olivotto et al.	2011	21835320	G+=42 G-=19	13/42	3/19	p=0.34	low/poor
Tobita et al.	2018	29386531	G+=24 G-=17	3/24	3/17	p=0.679	low/poor
Jääskeläinen et al.	2019	30775854	G+= 255 G-=227	33/255	8/227	p<0.001	fair
Pediatric							
Loar et al.	2015	26061417	G+=71 G-=66	22/71	16/66	NS	low/poor
Morita et al.	2008	18403758	G+=46 G-=38	7/46	3/38	NR	low/poor

ST8 - Geno Pheno ICD (2 of 2)

MYBPC3* v. *MYH7				**ICD**			
Author	**Year**	**PMID**	**N**	***MYBPC3***	***MYH7***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult							
Bongini et al.	2016	26869393	MYBPC3= 137 MYH7= 67	30/137	15/67	p=0.94	low/poor
Curila et al.	2012	22455086	MYBPC3= 21 MYH7=11	3/21	0/11	p=0.1999	low/poor
Waldmuller et al.	2011	21750094	MYBPC3=52 MYH7=35	15/52	11/35	NS	low/poor
Tobita et al.	2018	29386531	MYBPC3=11 MYH7=12	2/11	2/12	p=1.0	low/poor
Miller et al.	2019	31199839	MYBPC3=37 MYH7=41	20/37	11/41	p=0.020	good
Lee et al.	2018	30354366	MYBPC3=659 MYH7=296	53/659	23/296	NR	good

Multiple v. Single				**ICD**			
Author	**Year**	**PMID**	**N**	***Multiple***	***Single***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult							
Van Driest et al.	2004	15519027	Multiple=10 Single=137	4/10	33/137	p=0.26	low/poor
Wang et al.	2014	25132132	Multiple=38 Single=194	1/38	2/194	p=0.4168	fair
Pediatric							
Mathew et al.	2018	29053178	Multiple=16 Single=82	HR 4.42 (1.77-11.02)		p<0.001	fair

LEGEND

PMID = PubMed ID

G+ = genotype postiive cohort

G- = genotype negative cohort

ICD = implantable cardioverter defibrillator

NS = not significant

NR = not reported (when it could have been reported)

HR = hazard ratio

. = not applicable

ST9 - Geno Pheno Heart Failure (1 of 2)

Genotype+ v. Genotype-				NYHA class III/IV			Transplant/VAD			Other Heart Failure outcomes			
Author	**Year**	**PMID**	**N**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult													
Fujita et al.	2013	24621997	G+=78 G-=98	.	.	.	.	.	.	HF admission: NR	HF addmision: NR	p=0.172	Fair
Tran Vu et al.	2019	31308319	G+=44 G-=60	NYHA III: 1/44	NYHA III: 0/60	p>0.05	.	.	.	.	.	.	Low/Poor
Lopes et al.	2019	30554928	G+=122 G-=302	.	.	.	.	.	.	HF symptoms: 42/116	HF symptoms: 115/274	p=0.289	Poor
Neubauer et al.	2019	31699273	G+=943 G-=1,693	.	.	.	.	.	.	LVEF <55%: 126/882	LVEF <55%: 227/1,598	p=0.983 (unadjusted) p=1.000 (Bonferroni adjusted)	Good
Agarwal et al.	2015	26271555	G+=57 G-=67	NYHA III/IV: 10/57	NYHA III/IV: 17/67	p=0.2898	.	.	.	.	.	.	Fair
Chiou et al.	2015	25086479	G+=13 G-=25	NYHA III/IV: 8/13	NYHA III/IV: 14/25	NR	.	.	.	.	.	.	Very Low/Very Poor
Gruner et al.	2013	23239831	G+=163 G-=308	NYHA III/IV 23/163	NYHA III/IV: 41/308	p=0.8097	.	.	.	.	.	.	Low/Poor
Li et al.	2014	24909666	G+=198 G-=360	NYHA III/IV: 19/198	NYHA III/IV: 25/360	p=0.259	.	.	.	HF event: HR 4.51 (2.09-9.31)		p<0.001	Low/Poor
Towe et al.	2015	25495007	G+=18 G-=53	NYHA III/IV: 6/17	NYHA III/IV: 10/51	p=0.70	.	.	.	.	.	.	Very Low/Very Poor
van Velzen et al.	2016	27476098	G+=234 G-=278	NYHAII/III/IV: 81/234	NYHAII/III/IV: 121/278	<0.001	4/234	2/278	p=0.300	.	.	.	Good
Tobita et al.	2018	29386531	G+=24 G-=17	NYHAIII/IV: 10/24	NYHAIII/IV: 4/17	p=0.125	2/24	3/17	p=0.633	LVEF <50%: 18/24	LVEF<50 %: 6/17	p=0.023	Low/Poor
Pediatric													
Morita et al.	2008	18403758	G+=46G-=38	.	.	.	4/46	1/38	NR	.	.	.	Low/Poor

MYBPC3* v. *MYH7				NYHA class III/IV			Transplant/VAD			Other Heart Failure outcomes			
Author	**Year**	**PMID**	**N**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult													
Li et al.	2014	24909666	MYBPC3+=88 MYH7+=74	.	.	.	.	.	.	HF complications: 13/88	HF complications: 12/74	p=0.949	Low/Poor
Waldmuller et al.	2011	21750094	MYBPC3+=52 MYH7+=35	.	.	.	0/52	1/35	NS	.	.	.	Low/Poor
Robyns et al.	2020	31513939	MYBPC3+=80 MYH7+=27	.	.	.	.	.	.	HF episode: 16/80	HF episode: 7/27	NR	Low/Poor
Lopes et al.	2019	30554928	MYBPC3+=62 MYH7+=37	.	.	.	.	.	.	HF symptoms: 20/60	HF symptoms: 19/36	NR	Poor
Ho et al.	2018	30297972	MYBPC3+=745 MYH7+=377		NYHA III/IV: HR 1.94 (1.41-2.66)	NR		HR 2.76 (1.31-5.83)	NR		LVEF <55%: HR 0.52 (0.06-4.70) LVEF <35%: HR 0.52 (0.23-1.21)	NR NR	Good
Biagini et al.	2014	25037680	MYBPC3+=38 MYH7+=26	NYHA III/IV: 24/38	NYHA III/IV: 19/26	NR	.	.	.	.	.	.	Fair
Bongini et al.	2016	26869393	MYBPC3+= 137 MYH7+= 67	NYHA III/IV: 20/137	NYHA III/IV: 6/67	p=0.259	.	.	.	HF hospitalization: 16/137	HF hospitalization: 5/67	p=0.35	Low/Poor
Chida et al.	2017	27885498	MYBPC3+=19 MYH7+=31	NYHA III/IV: 2/19	NYHA III/IV: 3/31	NR	0/19	0/31	NR	.	.	.	Very Low/Very Poor
Tobita et al.	2018	29386531	MYBPC3+=11 MYH7+=12	NYHA III/IV: 5/11	NYHA III/IV: 7/12	p=0.684	1/11	1/12	p=1.0	LVEF <50%: 8/11	LVEF <50%: 8/12	p=1.0	Low/Poor

ST9 - Geno Pheno Heart Failure (2 of 2)

Multiple v. Single				NYHA class III/IV			Transplant/VAD			Other Heart Failure outcomes			
Author	**Year**	**PMID**	**N**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult													
Zou et al.	2013	23283745	Multiple=19 Single=83	.	.	.	.	.	.	LVEF: NR	LVEF: NR	NS	Low/Poor
Biagini et al.	2014	25037680	Multiple=21 Single=83	NYHA III/IV: 13/21	NYHA III/IV: 55/83	NR		.	.	.	.	.	Fair
Fourey et al.	2017	28420666	Multiple=24 Single=140	.	.	.	.	.	.	EF <50%: 2/24	EF <50%: 2/140	p=0.10	Low/Poor
Ho et al.	2018	30297972	.	NYHA III/IV: HR 0.81 (0.35-1.87)		NR	HR 7.48 (2.73-20.47)		NR	LVEF <55%: HR 7.12 (0.78-64.60) LVEF <35%: HR 2.16 (0.60-7.71) HF, EF>55%: HR 1.82 (0.72-4.60)		NR NR NR	Good
Pediatric													
Mathew et al.	2018	29053178	Multiple=16 Single=82	.	.	.	HR ∞ (0-∞)	p=1.0	.	.	.	.	Fair

LEGEND
PMID = PubMed ID
G+ = genotype postiive cohort
G- = genotype negative cohort
NYHA = New York Heart Association
VAD = ventricular assist device
HR = heart failure
LVEF = left ventricular ejection fraction
EF = ejection fraction
NS = not significant
NR = not reported (when it could have been reported)
HR = hazard ratio
. = not applicable

ST10 - Geno Pheno Septal Reduct (1 of 2)

Genotype+ v. Genotype-				Myectomy			Ablation			Combined Myectomy/Ablation			Apical myectomy			
Author	**Year**	**PMID**	**N**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																
Agarwal et al.	2015	26271555	G+=57 G-=67	6/57	6/67	p=0.76	6/57	12/67	p=0.25	.	.	.	.	.	.	fair
Chiou et al.	2015	25086479	G+=13 G-=25	.	.	.	4/13	13/25	p=0.036	.	.	.	.	.	.	very low/very poor
Li et al.	2014	24909666	G+=198 G-=360	44/198	65/360	p=0.30	6/198	9/360	p=0.69	50/198	72/360	p=0.22	.	.	.	low/poor
Towe et al.	2015	25495007	G+=18 G-=53	6/17	12/51	p=0.3	0/17	1/51	p=1.0	.	.	.	4/17	12/51	p=0.6	very low/very poor
Van Driest et al.	2004	15519027	G+=147 G-=242	65/147	95/242	p=0.33	.	.	.	.	.	.	.	.	.	low/poor
van Velzen et al.	2016	27476098	G+=234 G-=278	53/234	66/278	p=0.771	21/234	32/278	p=0.348	73/234	91/278	p=0.710	.	.	.	good
Lopes et al.	2019	30554928	NR	3/24	15/60	p=0.207	5/122	8/296	p=0.534	.	.	.	.	.	.	poor
Pediatric																
Loar et al.	2015	26061417	G+=71 G-=66	37/71	33/66	NS	.	.	.	.	.	.	.	.	.	low/poor

MYBPC3* v. *MYH7				Myectomy			Ablation			Combined Myectomy/Ablation			Apical myectomy			
Author	**Year**	**PMID**	**N**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																
Bongini et al.	2016	26869393	MYBPC3=137 MYH7=67	18/137	13/67	NS	16/137	7/67	NS	.	.	.	.	.	.	low/poor
Curila et al.	2012	22455086	MYBPC3=21 MYH7=11	.	.	.	9/21	3/11		.	.	.	.	.	.	low/poor
Waldmuller et al.	2011	21750094	MYBPC3=52 MYH7=35	7/52	3/35	NS	10/52	5/35	NS	.	.	.	.	.	.	low/poor
Wang et al.	2008	18383048	MYBPC3=18 MYH7=52	0/18	3/52	NR	0/18	1/52	NR	.	.	.	.	.	.	low/poor
Weissler-Snir et al.	2017	28193612	MYBPC3=75 MYH7=53	.	.	.	.	.	.	4/75	7/53	NS	.	.	.	low/poor
Miller et al.	2019	31199839	MYBPC3=37 MYH7=41							5/37	12/41	NS				good
Lopes et al.	2019	30554928	MYBPC3=91 MYH7=52	3/20	2/17	NR	3/91	3/52	NR	.	.	.	.	.	.	poor
Lee et al.	2018	30354366	MYBPC3=659 MYH7=296	.	.	.	.	.	.	52/659	17/296	NR	.	.	.	good

ST10 - Geno Pheno Septal Reduct (2 of 2)

Multiple v. Single				Myectomy			Ablation			Combined Myectomy/Ablation			Apical myectomy			
Author	**Year**	**PMID**	**N**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																
Van Driest et al.	2004	15519027	Multiple=10 Single=137	6/10	59/137	p=0.30	.	.	.	.	.	.	.	.	.	low/poor
Wang et al.	2014	25132132	Multiple=38 Single= 194	NR	NR	NS	NR	NR	NS	NR	NR	NS	.	.	.	fair
Fourey et al.	2017	28420666	Multiple=24 Single=140	.	.	.	.	.	.	2/24	29 /140	p=.26	.	.	.	low/poor
Pediatric																
Mathew et al.	2018	29053178	Multiple=16, Single=82	HR 0 (0-∞)	.	p=1	.	.	.	.	.	.	.	.	.	fair

LEGEND

PMID = PubMed ID
G+ = genotype postiive cohort
G- = genotype negative cohort
NS = not significant
NR = not reported (when it could have been reported)
HR = hazard ratio
. = not applicable

ST11 - Geno Pheno Mortality (1 of 1)

Genotype+ v. Genotype-					Death		All cause mortality			Cardiac mortality			HCM-related death			HF mortality			Survival			
Author	**Year**	**PMID**	**N**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	***G+***	***G-***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																						
Lopes et al.	2015	25351510	G+=383 G-=491	.	.	.	.	.	.	From 1st evaluation: HR 2.81 (1.21-6.51) From birth: HR 3.99 (1.71-9.36)	.	p=0.012 p=0.01	.	.	.	.	.	.	.	.	.	good
Towe et al.	2015	25495007	G+=17 G-=51	4/17	10/51	p=0.7	.	.	.	0/17	2/51	p=0.2	.	.	.	.	.	.	.	.	.	very low/very poor
van Velzen et al.	2016	27476098	G+=234 G-=278	.	.	.	HR 1.9 (1.14-3.151)		p=0.014	HR 2.82 (1.49-5.36)	.	p=0.002	.	.	.	HR 6.33 (1.79-22.41)		p=0.004	.	.	.	good
Ntusi et al.	2016	27841901	G+=10 G-=25	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	HR=1.8 (0.4-8.9)	.	p=0.474	fair
Tobita et al.	2018	29386531	G+=24 G-=17	.	.	.	1/24	0/17	p=1.0	.	.	.	.	.	.	.	.	.	.	.	.	low/poor
van Velzen et al.	2018	30292335	G+=430 G-=380	.	.	.	HR 0.90 (0.61-1.32)	.	p=.57	HR 1.02 (0.63-1.67)	.	p=0.92	.	.	.	HR 1.05 (0.48-2.28)	.	p=0.91	.	.	.	low/poor
Biagini et al.	2014	25037680	G+=104 G-=52	.	.	.	.	.	.	.	.	.	46%	48%	p=0.253	.	.	.	.	.	.	fair

MYBPC3* v. *MYH7					Death		All cause mortality			Cardiac mortality			HCM-related death			HF mortality			Survival			
Author	**Year**	**PMID**	**N**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	***MYBPC3***	***MYH7***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																						
Bongini et al.	2016	26869393	MYBPC3=137 MYH7=67	.	.	.	.	.	.	.	.	.	.	.	.	4/137	2/67	NR	.	.	.	very low/very poor
Chida et al.	2017	27885498	MYBPC3=19 MYH7=31	4/19	2/31	NR	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	low/poor
Wang et al.	2008	18383048	MYBPC=18 MYH7=52	2/9	10/52	NS	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	low/poor
Tobita et al.	2018	29386531	MYBPC3=11 MYH7=12	.	.	.	0/11	1/12	p=1.0	.	.	.	.	.	.	.	.	.	.	.	.	low/poor
Robyns et al.	2020	31513939	MYBP3=80 MYH7=27	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	p=0.08	low/poor
Miller et al.	2019	31199839	MYBPC=37 MYH7=41	.	.	.	2/37	1/41	p=0.601	.	.	.	.	.	.	.	.	.	.	.	.	good
Ho et al.	2018	30297972	MYBPC3=745 MYH7=377	.	HR 0.90 (0.52-1.55)	NR	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	good

Multiple v. Single					Death		All cause mortality			Cardiac mortality			HCM-related death			HF mortality			Survival			
Author	**Year**	**PMID**	**N**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	***Multiple***	***Single***	**p-value**	**Quality Assessment**
Adult/Mixed Peds and Adult																						
Wang et al.	2014	25132132	Multiple=38 Single=194	.	.	.	HR 5.54 (2.51-12.23)	.	p=2 × 10−5	.	.	.	.	.	.	HR 5.39 (1.63–17.84)	.	p=0.006	.	.	.	fair
Biagini et al.	2014	25037680	Multiple=21 Single=83	.	.	.	.	.	.	NR	NR	NS	.	.	.	.	.	.	.	.	.	fair
Ho et al.	2018	30297972	Multiple=34 Single=1210	HR 0.50 (0.13-1.91)	.	NR	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	good
Pediatric																						
Mathew et al.	2018	29053178	Multiple=16 Single=82	HR 2.38 (0.22-26.42)	.	p=0.48	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	fair

LEGEND
PMID = PubMed ID
G+ = genotype postiive cohort
G- = genotype negative cohort
NS = not significant
NR = not reported (when it could have been reported)
HR = hazard ratio
. = not applicable

ST12 - Penetrance (1 of 3)

Author	Year	PMID	% Penetance	n	N	Probands Included	Genes/Variants Included	Average Age of At-Risk Family Members	Average Age-of-Onset	% dx by certain age	Quality Assessment
Watkins et al.	1995	7898523	76	51	67	N	TNNT2	.	.	.	Low/poor
Charron et al.	1997	9219008	Overall 68 MYBPC3 67 TNNT2 100 MYH7-67	Overall 62 MYBPC3 25 TNNT2 4 MYH7-33	Overall 90 MYBPC3-37 TNNT2-4 MYH7-49	N	MYPC3, TNNT2, MYH7	.	.	25% by 10 years (1/4) 53%% by 20 years (9/17) 51% by 30 years (20/39) 57% by 40 years (31/54) 61% by 50 years (43/70) 68% by 60 years (56/83) 68% by 70 years (59/87) Male vs Female 8/49 (77.5%) VS 58.5% (24/41) OR 3.98 p=0.011 (CI 1.34-11.48)	Low/poor
Niimura et al.	1998	9562578	57	121	212	N	MYBPC3	.	.	.	Very low/very poor
Blair et al.	2002	11861413	100	17	17	Y	MYH7	.	.	.	Low/poor
Havndrup et al.	2003	12566107	71	27	38	N	MYH7	.	.	.	Very low/very poor
Andersen et al.	2004	15114369	57	25	44	N	MYBPC3	.	.	.	Very low/very poor
Mogensen et al.	2004	15607392	48	48	100	Y	TNNI3	.	.	.	Fair
Perrot et al.	2005	15856146	65	11	17	Y	MYH7	.	.	95% over 30 years	Low/poor
Arad et al.	2005	16267253	89	16	18	N	ACTC1	.	.	.	Low/poor
Doolan et al.	2005	15698845	56	5	9	Y	TNNI3	.	.	.	Very low/very poor
Song et al.	2005	15563892	63	45	72	Y	MYBPC3, MYH7, TNNT2	MYH7-34.6+14.0 MYBPC3-48.4+13.5 TNNT2-31.5+6.6 for	34.6+14.0 for MYH7 48.4+13.5 for MYBPC3 31.5+6.6 for TNNT2	.	Low/poor
Monserrat et al.	2007	17611253	100	42	42	Y	ACTC1	.	.	.	Low
Ehlermann et al.	2008	18957093	67	16	24	N	MYBPC3	.	.	.	Low/poor
Andersen et al.	2009	19035361	70	54	77	N	MYH7, MYBPC3, MYL3, TNNI3, TNNT2, CSRP3	.	.	.	Very low/very poor
Wang et al.	2008	18383048	20	9	46	N	MYBPC3, MYH7	.	MYH7 mutation +: 34.6 ± 14.0 MYBPC3 mutation +: 50.0 ± 15.0 (average age at presentation)	.	Very low/very poor
Michels et al.	2009	19666645	41	31	76	N	MYBPC3, MYH7, TNNT2, TPM1	40+/-14 years	.	.	Low/poor
Oliva-Sandoval et al.	2010	21088121	61.5	40	65	N	MYBPC3	37+/-18y,	41.3	50% by 46 years	Fair
Rodríguez-García et al.	2010	20433692	58	14	24	N	MYBPC3	.	.	.	Fair
Saltzman et al.	2010	20378854	58	45	77	Y	MYBPC3	.	.	.	Very low/very poor
Garcia-Pavia et al.	2011	21896538	69	9	13	N	MYH7, MYBPC3, MYL2, TNNI3, MYL3	.	.	.	Very low/very poor
Christiaans et al.	2011	21459882	30	136	446	N	MYBPC3, MYH7, TNNT2, TPM1, MYL2	39.3+17.6 (range 1–86)	61.6 (median) CI: 57.4-64.7	.	Fair
Gray et al.	2011	21862152	3.1	1	32	N	MYBPC3, MYH7, TNNI3, ACTN2	23+/-15 at initial visit 16>18yo, 16</+18yo	.	.	Low/poor
Brito et al.	2012	22857948	75	77	102	N	MYBPC3, MYH7, TNNT2, TNNI3	.	.	.	Low/poor
Pasquale et al.	2012	22144547	76	52	68	Y	TNNT2	.	.	.	Fair
Miller et al.	2013	23054336	60	31	52 (HCM &DCM)	N	MYH7, MYPBC3, TNNT2, LMNA, ACTC, MYL3	.	.	.	Very low/very poor

ST12 - Penetrance (2 of 3)

Author	Year	PMID	% Penetance	n	N	Probands Included	Genes/Variants Included	Average Age of At-Risk Family Members	Average Age-of-Onset	% dx by certain age	Quality Assessment
Jensen et al.	2013	23197161	17	2	12	N	MYH7, MYBPC3, MYL2	22+/-6 (unknown genetic status) 23+/-5 (G+ status) 22+/-4 (G- controls)	.	.	Low/poor
Jääskeläinen et al.	2013	22462493	Overall - 64 (without probands) TPM1-71 (without probands) MYPBC3-58 (without probands) Overall = 80 (with probands) MYBPC3 =78 (with probands)	Overall - 44 (without probands) TPM1-22 (without probands) MYBPC3-22 (without probands) Overall = 99 (with probands) MYBPC3 =57 (with probands)	Overall - 69 (without probands) TPM1-31 (without probands) MYPBC3- 38 (without probands) Overall = 124 (with probands) MYBPC3 =73 (with probands)	N/Y	TPM1 (D175N) MYBPC3 (Q1061X)	.	.	.	Very low/very poor
Das et al.	2014	24113344	62	34	55	N	MYPBC3, MYH7, TNNI3, ACTN2, TNNT2, MYL3, PLN	.	.	.	Very low/very poor
Valdés-Mas et al.	2014	25351925	76	16	21	Y	FLNC	.	.	87% by 40 years of age	Very low/very poor
Calore et al.	2015	25740977	64	29	45	N	MYBPC3	.	44 +/- 19	77% of men dx by 40 years 35% of women by 40 years	Fair
Claes et al.	2016	26497160	74	28	38	N	MYL2	.	53	.	Low/poor
Terauchi et al.	2015	25123604	85	50	61	Y	MYBPC3	.	47	92% in males by 40 years 67% in females by 40 years	Low/poor
Ripoll-Vera et al.	2016	26507537	77	37	48	Y	TNNT2	.	37	50% by 37 years 75% at 50 years	Very low/very poor
Bales et al.	2016	26936621	81	64	79	Y	MYH7, MYBPC3, TNNT2, TNNI3, TPM1, MYL2	.	.	.	Very low/very poor
Mattos et al.	2016	27737317	39 (without probands) 56 (with probands)	7 (without probands) 14 (with probands)	18 (without probands) 25 (with probands)	N/Y	MYBPC3, MYH7	Pheno+ 42 ± 20 Pheno- 32 ± 17	.	.	Fair
Sabater-Molina et al.	2017	28029522	72	39	54	N	MYBPC3	42.0+/-20.1	44	50% by 44 years	Fair
Vermeer et al.	2017	28396031	7	8	119	N	MYBPC3, MYH7, TNNT2	12.1 ± 3.4 years	17	4% by 18 years	Fair
Zhao et al.	2017	28450932	44	18	41	N	MYBPC3	G+/LVH+ = 48 + 12; G+/LVH- = 38 + 10	.	.	Fair
Alejandra Restrepo-Cordoba et al.	2017	28138913	47	27	57	N	.	.	.	.	Very low/very poor
Viswanathan et al.	2017	29121657	70	23	33	N	ACTC1, GLA, JPH2, LAMP2, LMNA, MYBPC3, MYH7, MYL3, PRKAG2, TNNC1, TNNI3, TNNT2, TPM1, TTR	Pheno+ 42 +/-2 Pheno- 43 +/-4	.	.	Very low/very poor

ST12 - Penetrance (3 of 3)

Author	Year	PMID	% Penetance	n	N	Probands Included	Genes/Variants Included	Average Age of At-Risk Family Members	Average Age-of-Onset	% dx by certain age	Quality Assessment
van Velzen et al.	2017	28794111	43 (without probands) 72 (with probands)	56 (without probands) 188 (with probands)	130 (without probands) 262 (with probands)	N/Y	MYBPC3	relatives 42 +/- 15 years probands 45 +/- 14	.	.	Low/poor
van Velzen et al.	2018	29661763	46 (without probands) 66 (with probands	122 (without probands) 271 (with probands)	264 (without proband) 413 (with probands)	N/Y	<2012: MYBPC3, MYH7, MYL2, MYL3, TNNT2, TNNI3, CSRP3, TCAP, TPM1, ACTC1, and TNNC1 >2012 MYBPC3, MYH7, TNNT2. TNNI3, MYL2, MYL3, FHL1, TPM1, ALPK3, MIB1	41 +/-18	.	14% (6/41) <12 years 44% (43/98) >50 years	Fair
Jalanko et al.	2018	30497761	70	98	140	Y	MYBPC3 (Gln1061X) TPM1 (Asp175Asn)	Pheno+ 48+/- 14 Pheno- 33+/-15	.	.	Very low/very poor
Mathew et al.	2018	29053178	Overall -61 MYH7- 76 Multiple variants - 81	Overall - 60 MYH7 - 26 Multiple variants- 13	Overall - 98 MYH7- 29 Multiple variants- 16	Y	MBPC3, MYH7, TPM1, MYL2, TTN, TNNI3, TNNT2, CSRP3	MYH7- 15.3 (9.8, 17.4) Other genes- 13.3 (7.8, 17.2)	12.4 (IQR 5.7-15.2) overall 9.0 (IQR 5.2-14.1) for MYH7 13.0 (IQO 7.0-15.8) for other genes 8.9 (IQR 3.1-11.2) for multiple variants 13.1 (IQR 6.9-15.8) for single variant	.	Very low/very poor
Maurizi et al.	2019	30715937	10	21	203	N	ACTN2, MYBPC3, MYH7, MYL2, MYL3, TNNC1, TNNI1, TNNI3, TNNT2, TPM1	32+/-11 Pheno+ 30+/-14 Pheno- 34+/-12	.	.	Low/poor
Lorca et al.	2019	29631964	87	34	39	N	MYPBC3 p.G263*	Pheno+ 57 (no +/-)	.	97% in older than 40yrs and 100% in older than 55yrs	Low/poor
Lafreniere-Roula et al.	2019	31170284	15	80	524	N	MYH7, MYBPC3, PRKAG2, TPM1, TNNI, TNNT, MYL2, TMPO	13.4 (8.8-16.6)	.	.	Very low/very poor
Tarkiainen et al.	2019	30976029	68	32	47	Y	MYBPC3	Pheno+ 50+/- 11 Pheno- 33+/- 16	.	.	Very low/very poor
Vigneault et al.	2019	30561279	55	42	76	Y	.	Pheno+ 30+/- 13 Pheno- 21+/-8	.	.	Very low/very poor
Knight et al.	2020	31229680	54	26	48	N	.	.	.	.	Very low/very poor
Huurman et al.	2020	31776912	48	96	199	N	.	44+/- 14	.	.	Low/poor

LEGEND
PMID = PubMed ID
N = no
Y = yes
G+ = genotype postive cohort
IQR = interquartile range
Pheno+ = phenotype positive
. = not applicable

Supplementary Tables References

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